

Basic Principles of Medicine 1

Module: Foundations to Medicine

Lecture No: 15

Title: Structure and Functions of Proteins

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Objectives: DLA: Amino acids: Structure and function

- Before lecture 'Structure and function of proteins'

SOM.MK.I.BPM1.1.FTM.3.BCHM.0078	Classify amino acids based on their R groups (with examples for each group).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0079	Name the amino acids with a charged side chain, the amino acids containing a hydroxyl group and the amino acids containing sulphur.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0081	Discuss the role of histidine residues in buffering in hemoglobin.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0083	Discuss the importance of derivatives of amino acids (GABA, histamine, serotonin, nitric oxide, and catecholamines).

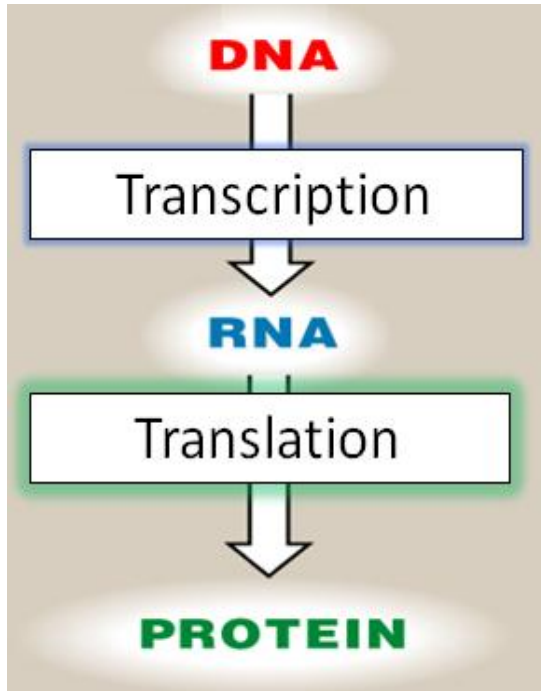
Recommended Reading

- **DLA:** Properties of amino acids
- **Lippincott's Illustrated Reviews Biochemistry, 9th Edition:**
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - **Chapter 2:** Protein Structure: [Protein Structure | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - **Concept map:** Figure 2.15
 - **Questions:** 2.1 – 2.7
- **Marks' Basic Medical Biochemistry:**
 - Chapter 7: [Structure–Function Relationships in Proteins | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library](#)

Objectives: Structure and function of proteins

SOM.MK.I.BPM1.1.FTM.3.BCHM.0087	Differentiate between the hierarchical levels of protein structure using the example of myoglobin and hemoglobin.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0088	Discuss the significance of bond forces involved in maintenance of protein structure (peptide bonds, disulfide bonds, hydrogen bonds, ionic bonds, hydrophobic forces and Van der Waals forces) using hemoglobin as an example.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0090	Relate that missense mutations can alter the solubility, function, charge and electrophoretic properties of proteins.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0091	Discuss neurochemical abnormalities in Prion disease and Alzheimer disease.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0092	Describe the role of chaperones and ubiquitin.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0093	Discuss protein denaturation in vivo and in the laboratory.

Proteins: Linear Chains of Amino Acids Linked Via Peptide Bonds

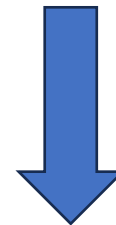


Sequence of amino acids in the **primary structure** is determined by the **genetic code**

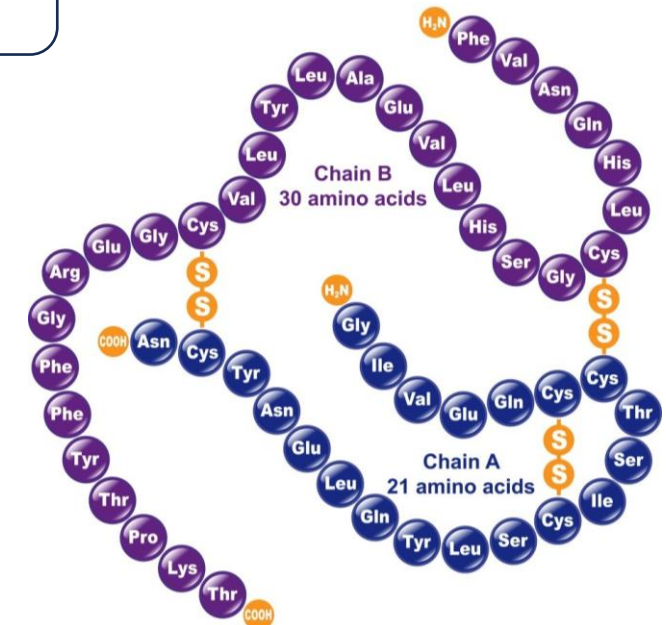


Post-translational modification
Protein Folding

Generates specific 3D structure



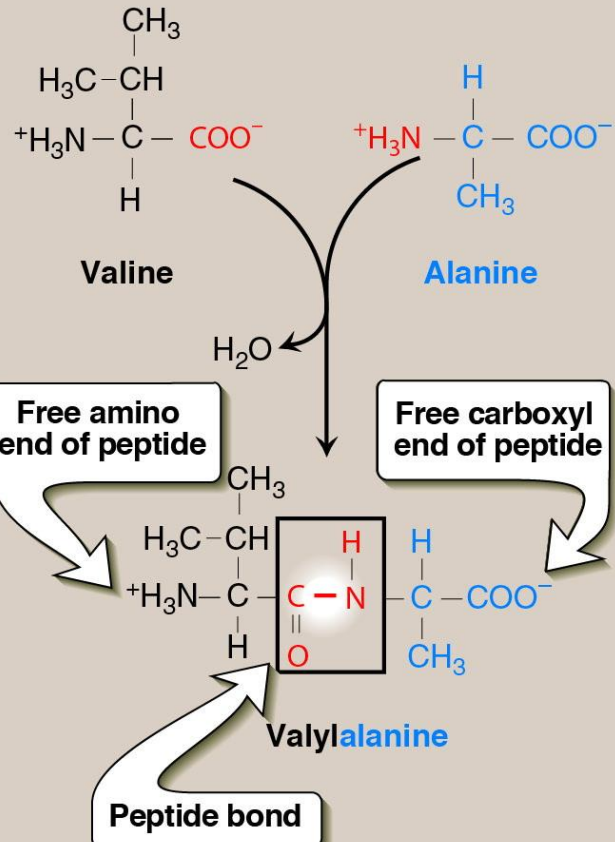
High diversity of functional proteins



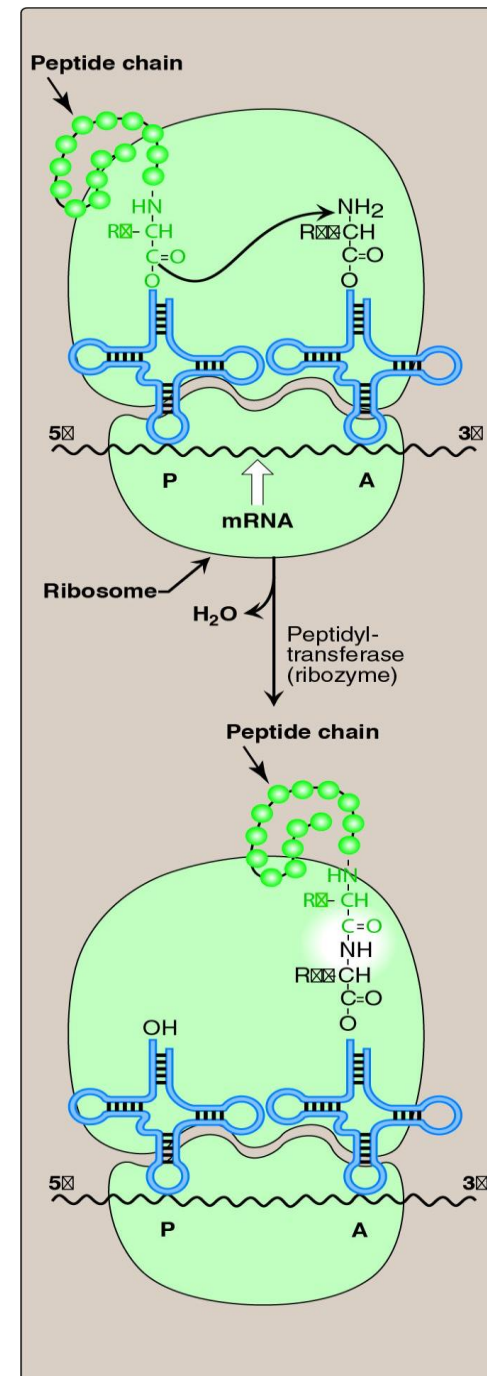
Human Insulin

Peptide bonds link amino acids

A Formation of the peptide bond

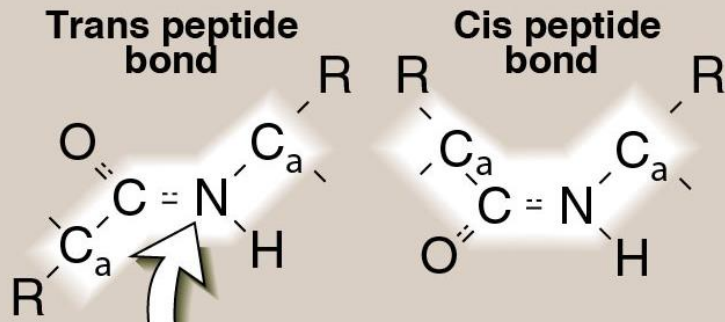


- In proteins, adjacent amino acids are joined covalently by peptide bonds
- Peptide bonds are formed between the α -carboxyl group of one amino acid and the α -amino group of the next amino acid
- Protein synthesis occurs on ribosomes (translation)
 - Uses t-RNA – binds amino acids
 - m-RNA (codons)



Characteristics of Peptide Bonds

B Characteristics of the peptide bond

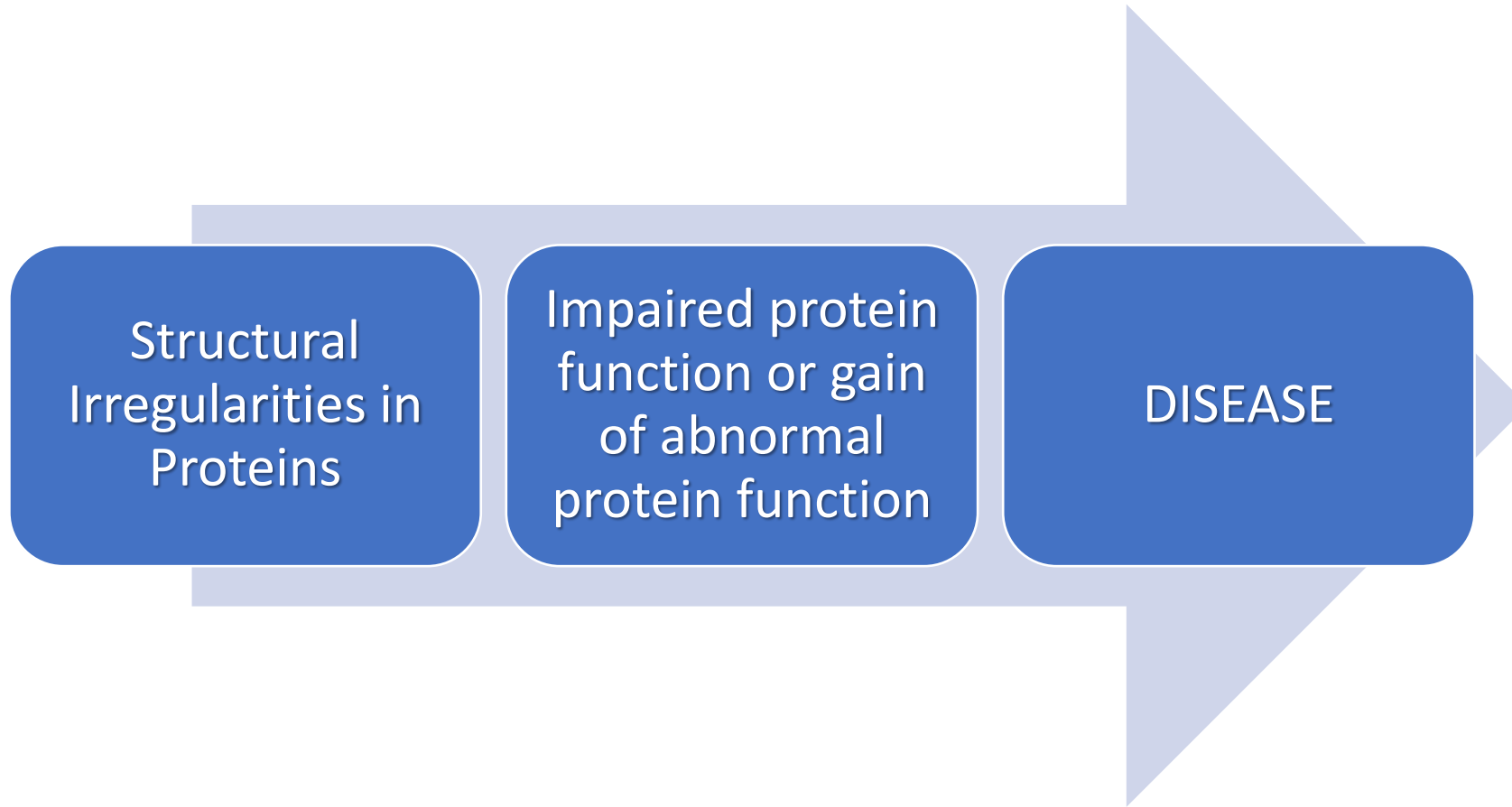


Peptide bonds in proteins

- Partial double-bond character
- Rigid and planar
- Trans configuration
- Uncharged but polar

- Has partial (40%) double-bond character
- Is rigid and planar
- Is uncharged but polar
- **Trans configuration**
- Side chains are in trans configuration
- The -C=O and -NH groups of the peptide bond may form hydrogen bonds
 - Stabilizes α -helices and β -sheets
- **Do not break during protein denaturation**

Why Understand Protein Structure?

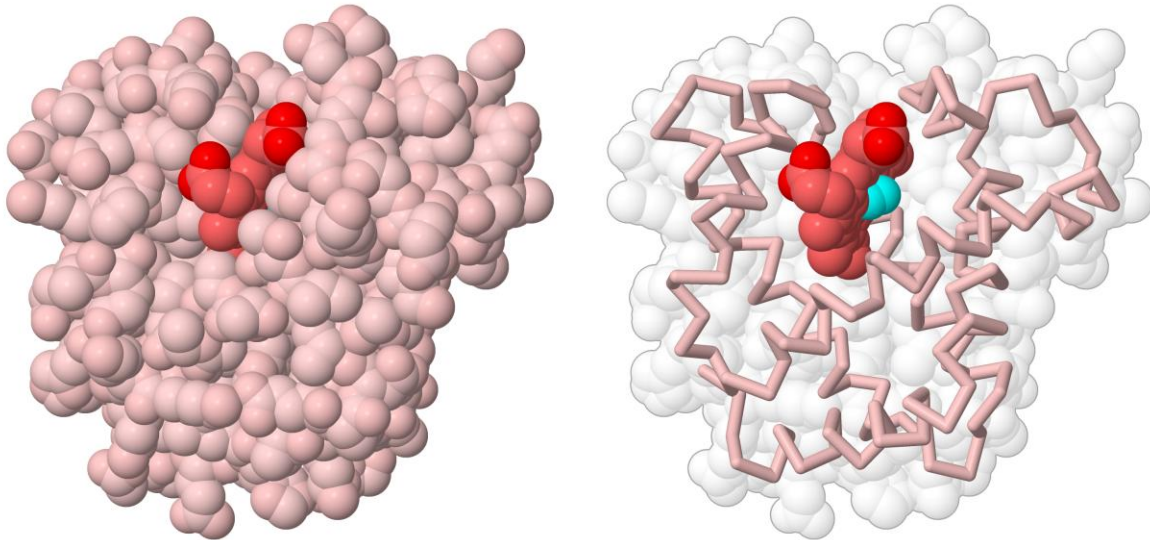


- Establish the link between structure and the physiological function of proteins
- Understand the molecular basis of disease
- Develop new targeted therapies

Structure & Function of Proteins are Related

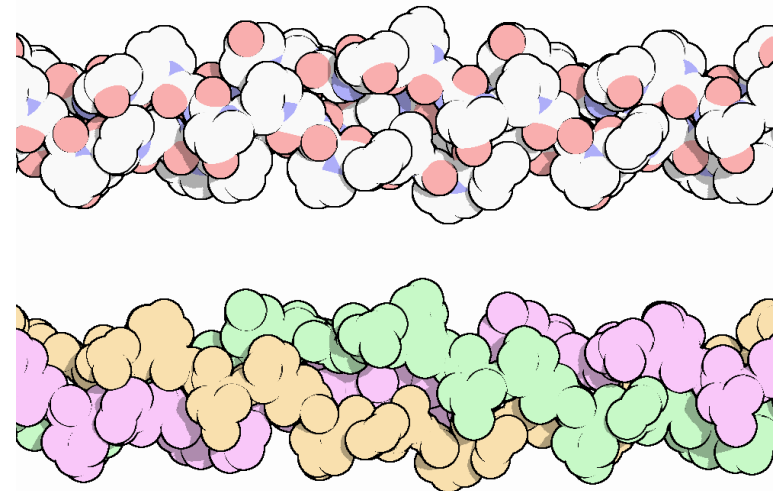
Proteins broadly classified into 2 groups:

1) Globular: Polypeptide chains folded into spherical or globular shapes



Myoglobin (pink) with oxygen (turquoise) bound to heme (red).

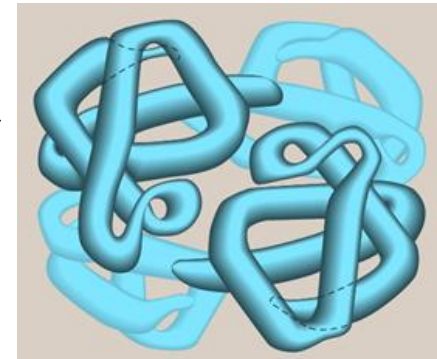
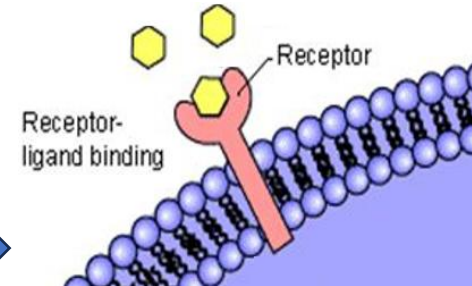
2) Fibrous: Polypeptide chains arranged in long strands



A portion of collagen, coloured by atom (top) & coloured to highlight the 3 chains (bottom).

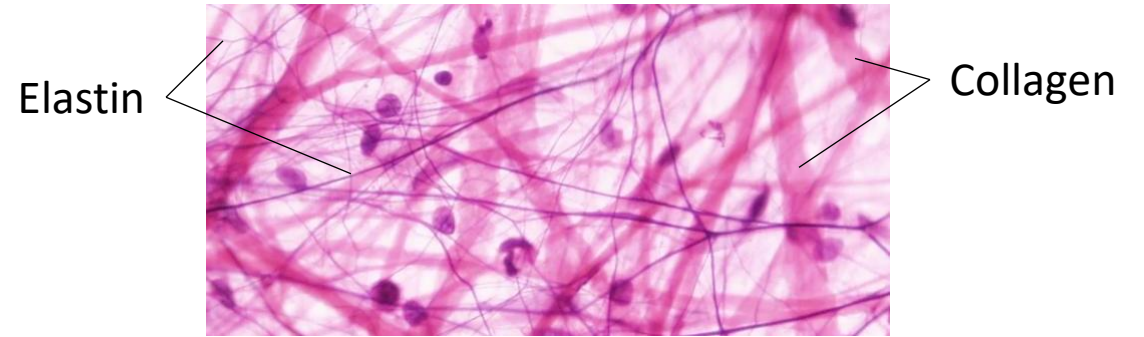
Some Functions of Globular Proteins

- Catalytic - **Enzymes** (can be regulated)
- Enzyme regulation
- Binding: to other proteins, ions & nucleic acids
- Regulatory: **E.g., Hormones; transcription factors**
- Defense: **E.g., Antibodies**
- Storage: **E.g., Ferritin**
- Signal Transduction
- Transport in blood: **E.g., Albumin**
- Oxygen delivery: **Hemoglobin**

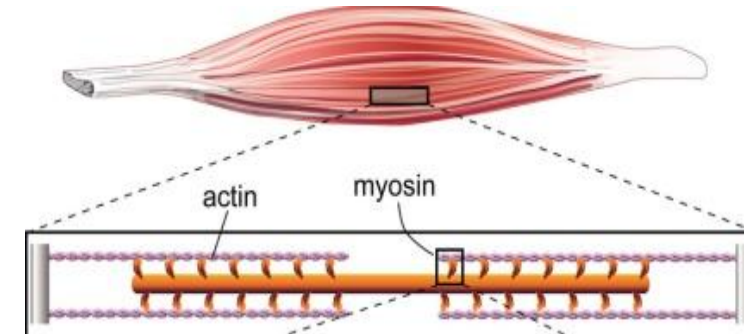


Some Functions of Fibrous Proteins

- **Extracellular Matrix:**
e.g., **Collagen, elastin**



- **Contractile:** **Actin and myosin**

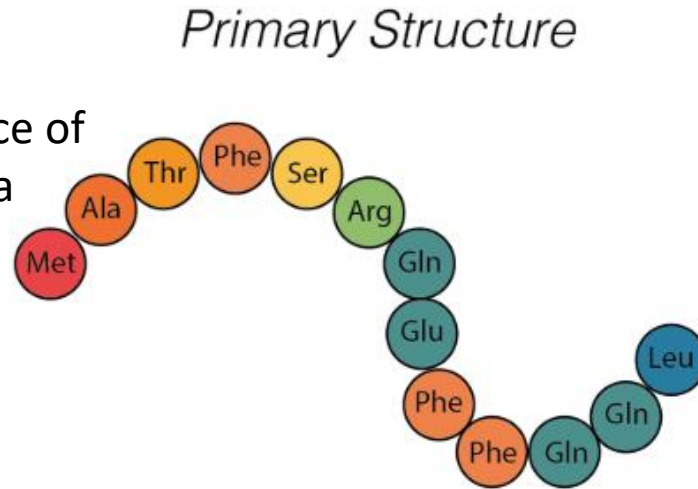


Four Levels of Protein Organization

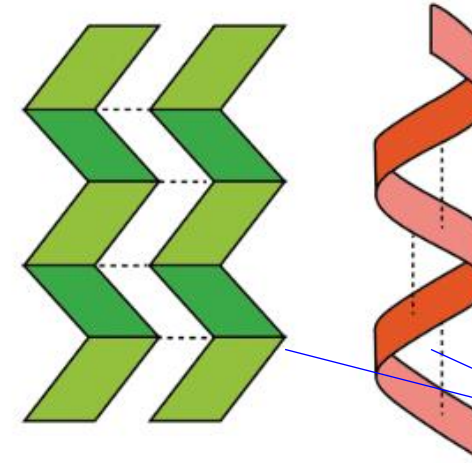
Self-study: Bonds stabilizing proteins

(Students are expected to review the information for better understanding)

1° Structure: Sequence of amino acids joined via peptide bonds



Secondary Structure



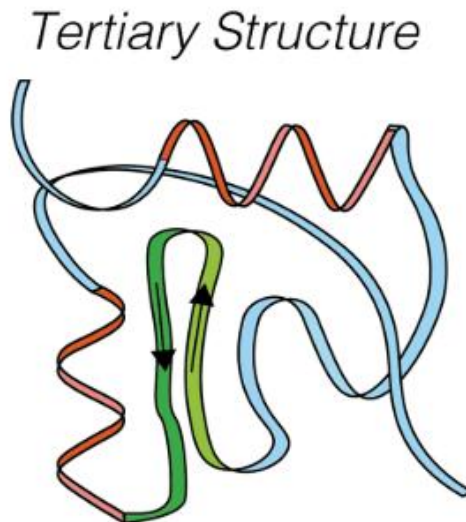
2° Structure: Elements which can be found repeatedly in the 3D structure of proteins

Stabilized by hydrogen bonds

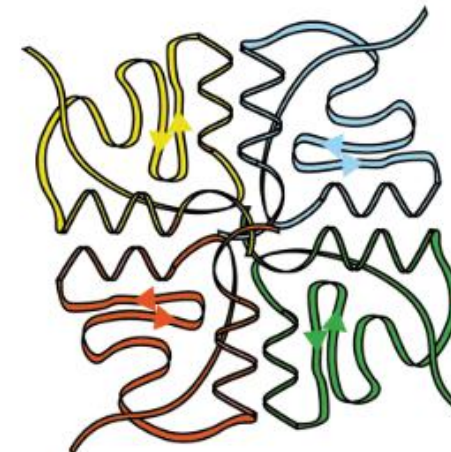
β -strand

α -helix

3° Structure: 3D folding of a protein that occurs due to amino acid side-chain interactions & core packing (includes functional domains)



Quaternary Structure



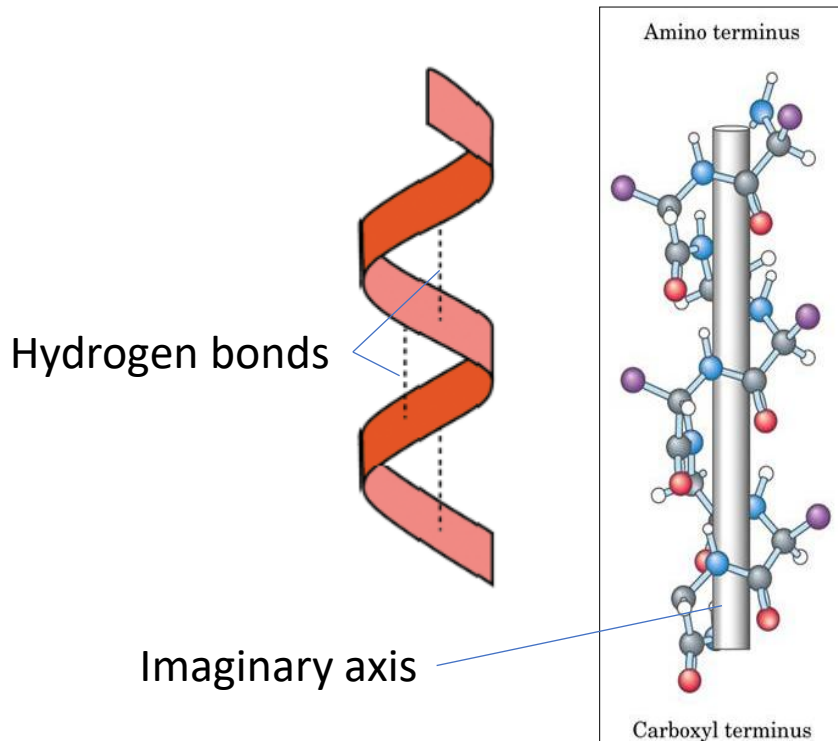
4° Structure: Association of a protein that consists of two or more polypeptide chains.

Stabilized by non-covalent bonds

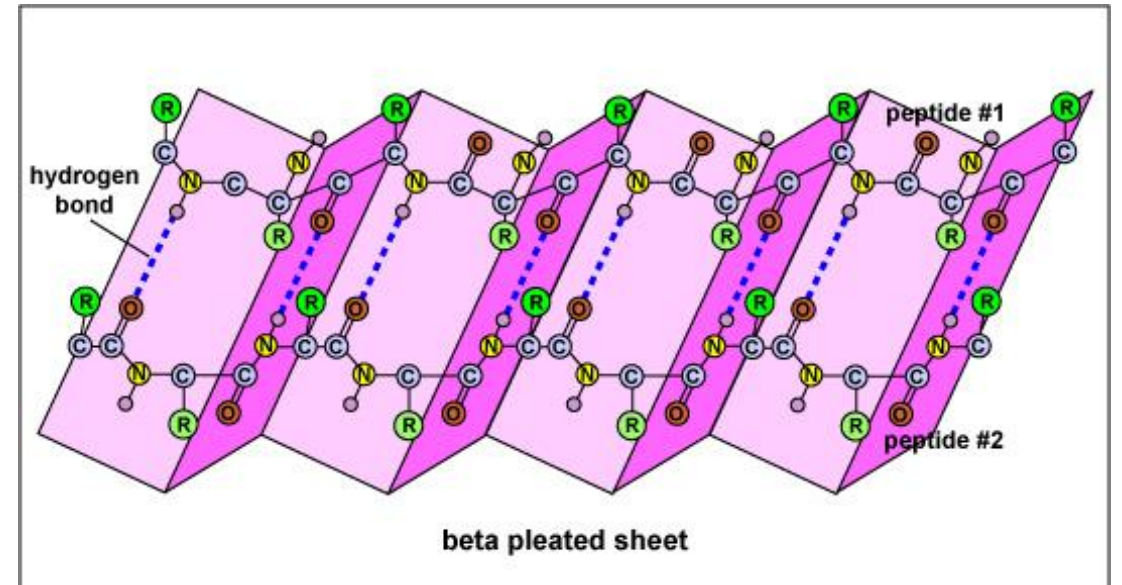
Secondary Structures

- Refers to the hydrogen bonding arrangement of the protein backbone.
- Local structures that are stabilized by hydrogen bonds.
- 2 types of secondary structure:

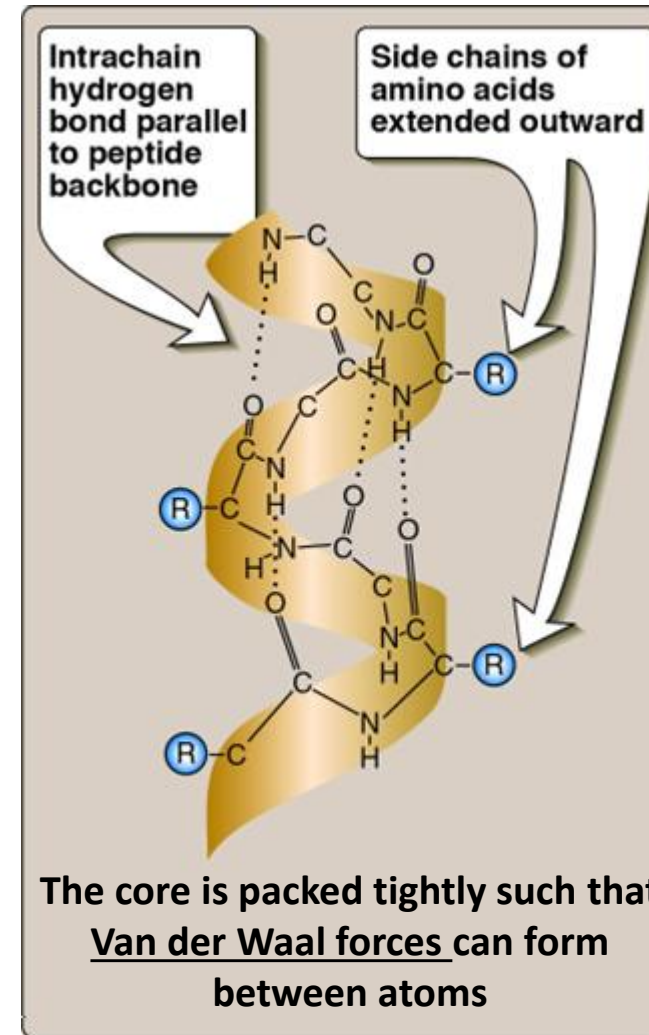
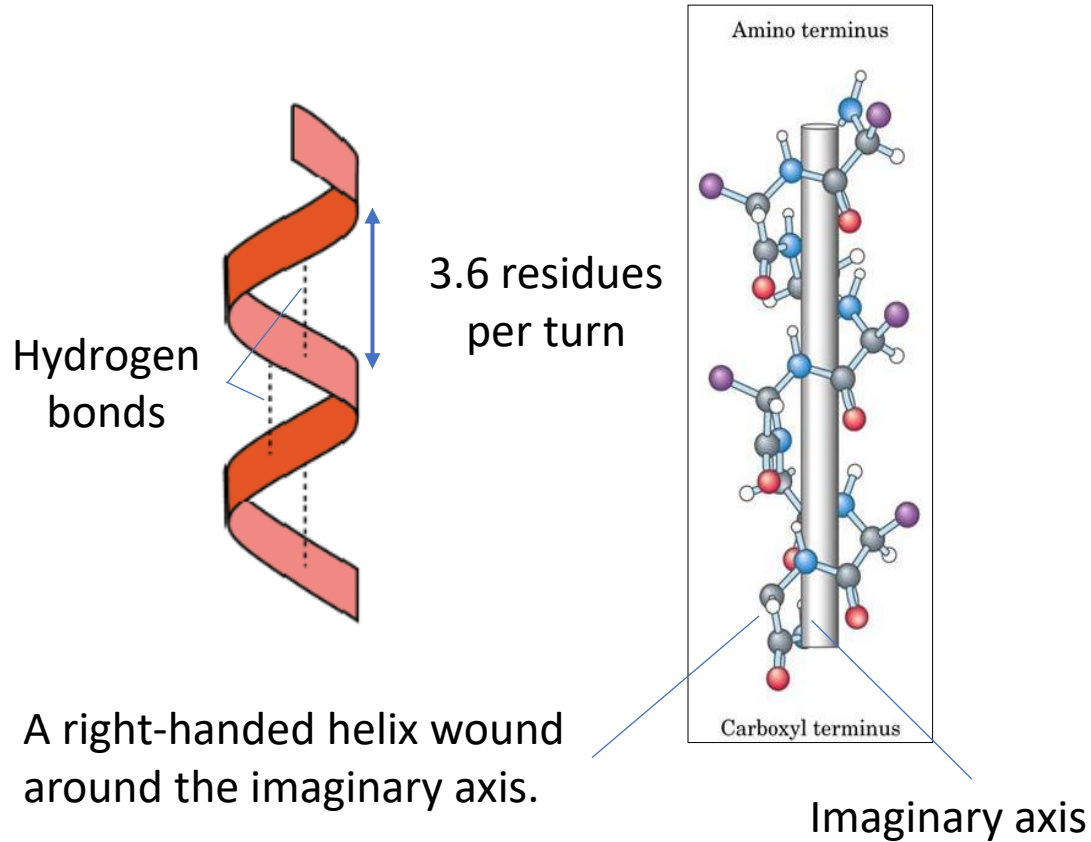
α -Helix



β -Pleated sheets



α -helix Stabilized by Hydrogen Bonds & Van der Waal Forces



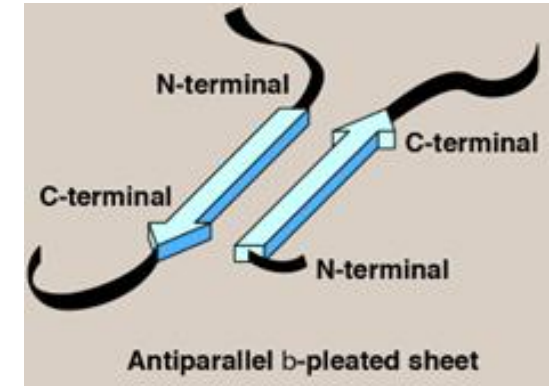
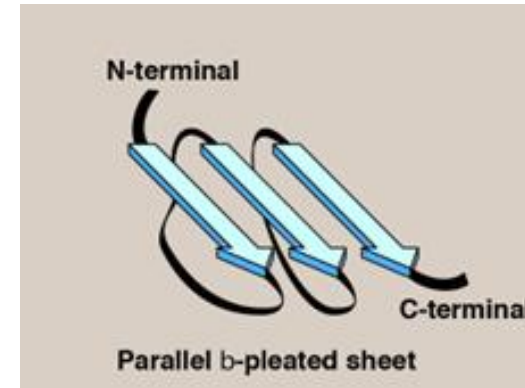
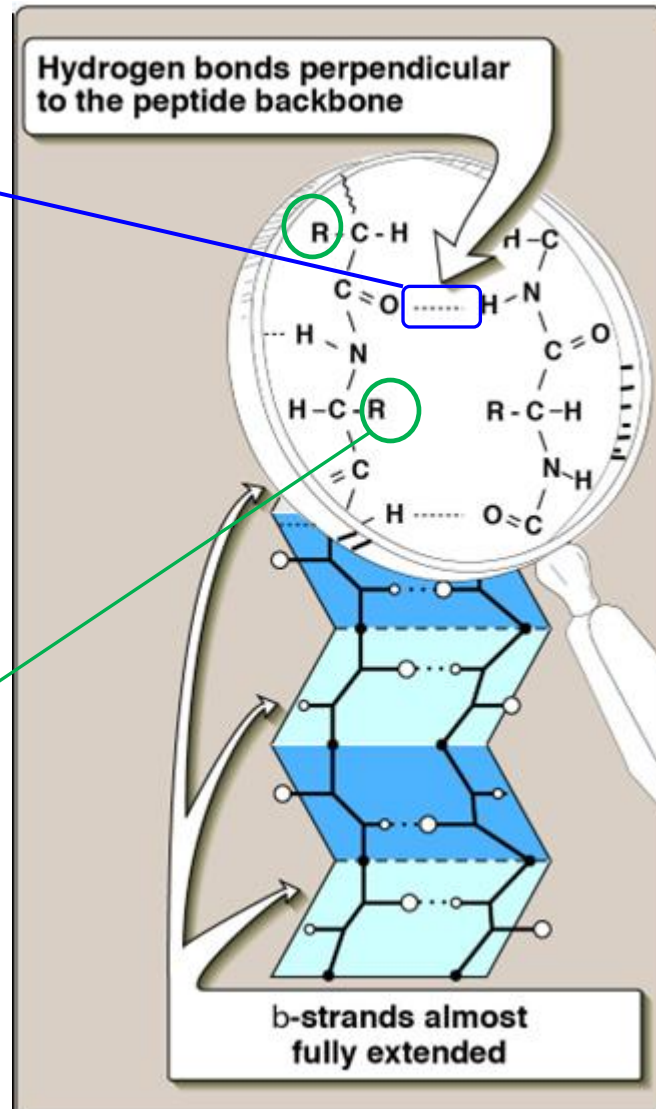
The α -helix is interrupted by electrostatic repulsion (or attraction) between charged amino acid side chains or due to steric hindrance by bulky side chains

● **Proline** or glycine residues often **terminate** the α -helix.

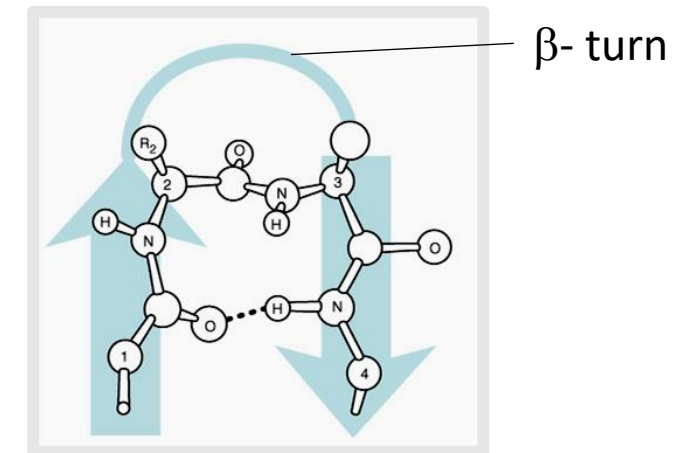
β -Pleated Sheet Stabilized by Hydrogen Bonds

H bonds between peptide bonds stabilize **β -pleated sheet** which is formed by two or more peptide chains

Amino acid side chains alternate above and below the plane of the pleated sheet



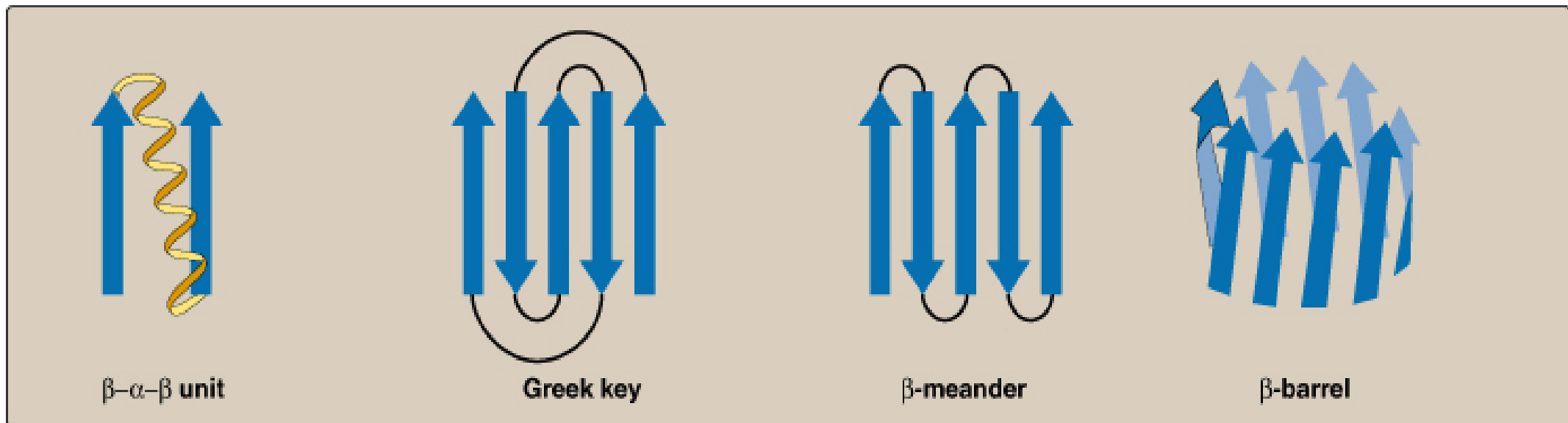
The protein strands can be anti-parallel or parallel



β -turn: Allows change in direction of the peptide chain

Super-secondary Structures

- **Super-secondary structures** (**motifs**) are formed when α - and β -strands combine through nonrepetitive links.
- Motifs are distinct and named according to shape or elements contained.

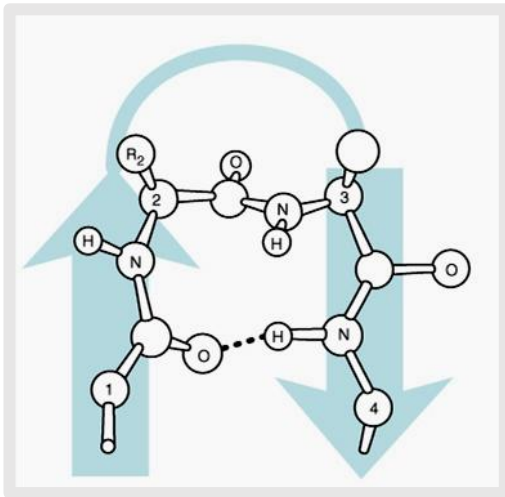


- Several motifs can combine to form **domains** which have specific functions

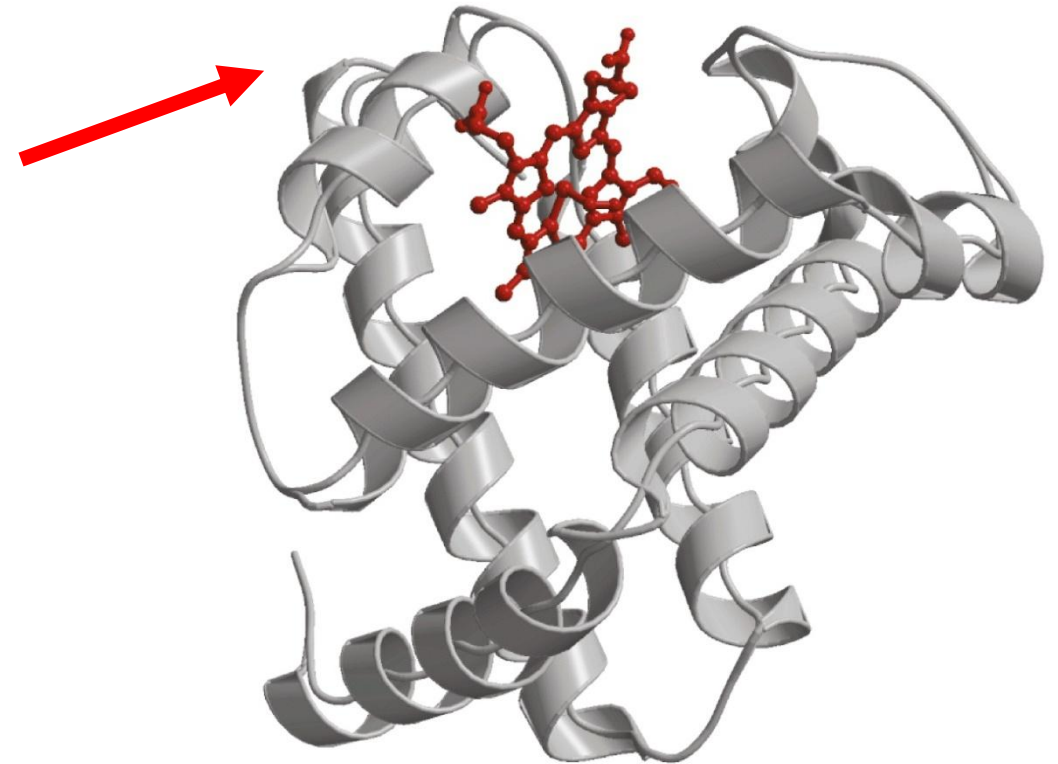
Non-Repetitive Regions of 2° Structure

Non-repetitive regions – link motifs

- More flexible than α - & β -structures



Proline is often used to introduce bend or kinks



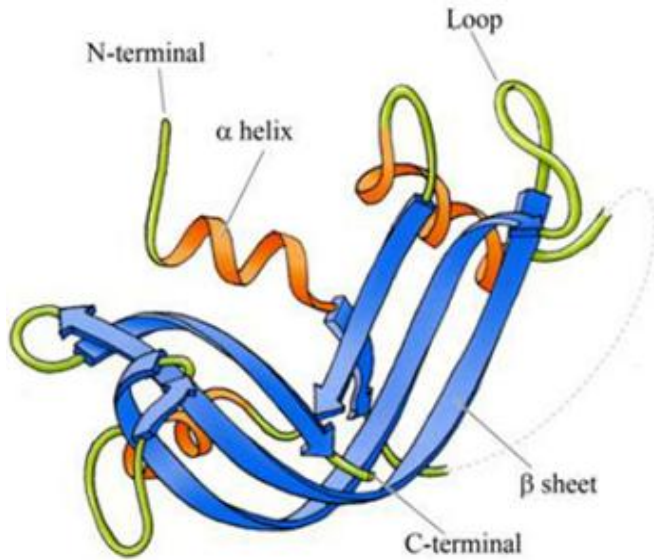
Myoglobin

Tertiary Structure of Proteins

- Describes the 3-D arrangement of all atoms in a protein in space due to **amino acid side-chain interactions**.
- Stabilized by **non-covalent interactions**

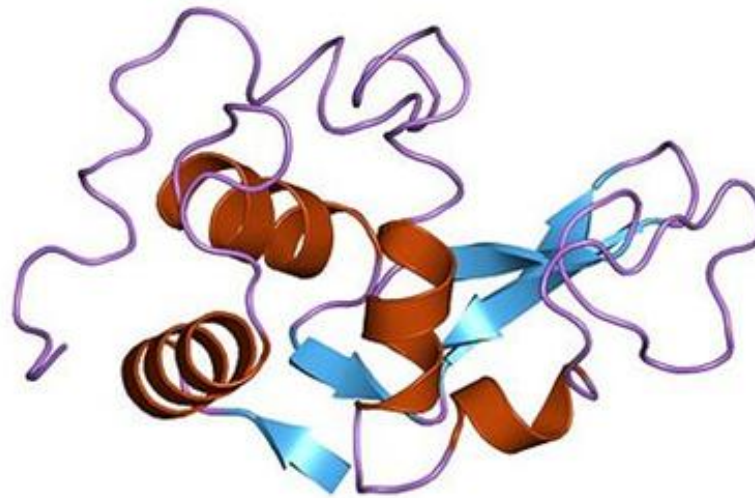
Self-study: Bonds stabilizing proteins

(Students are expected to review the information for better understanding)



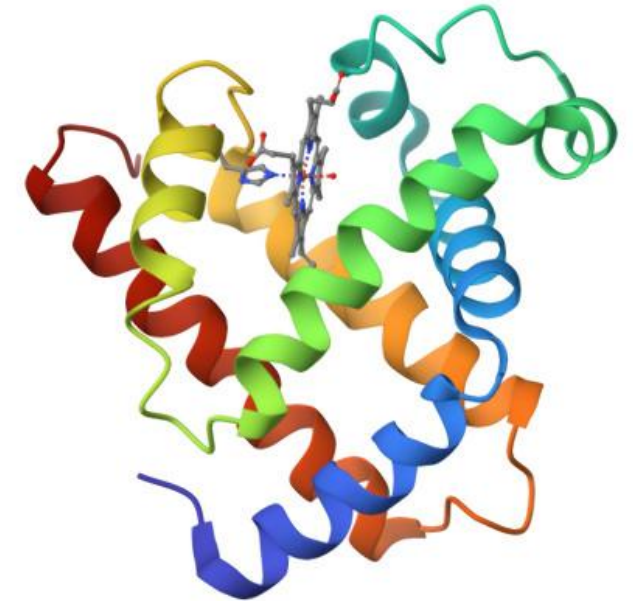
Ribonuclease

Juan M. Lara: Research gate



Lysozyme

Shyamala Mudavath: Research gate



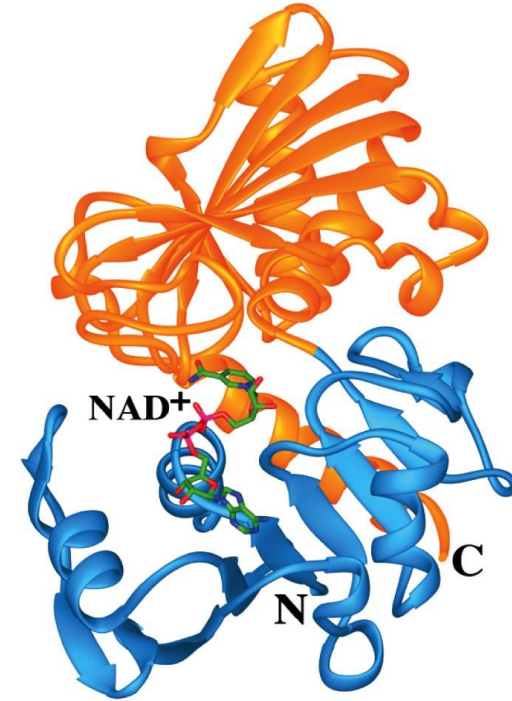
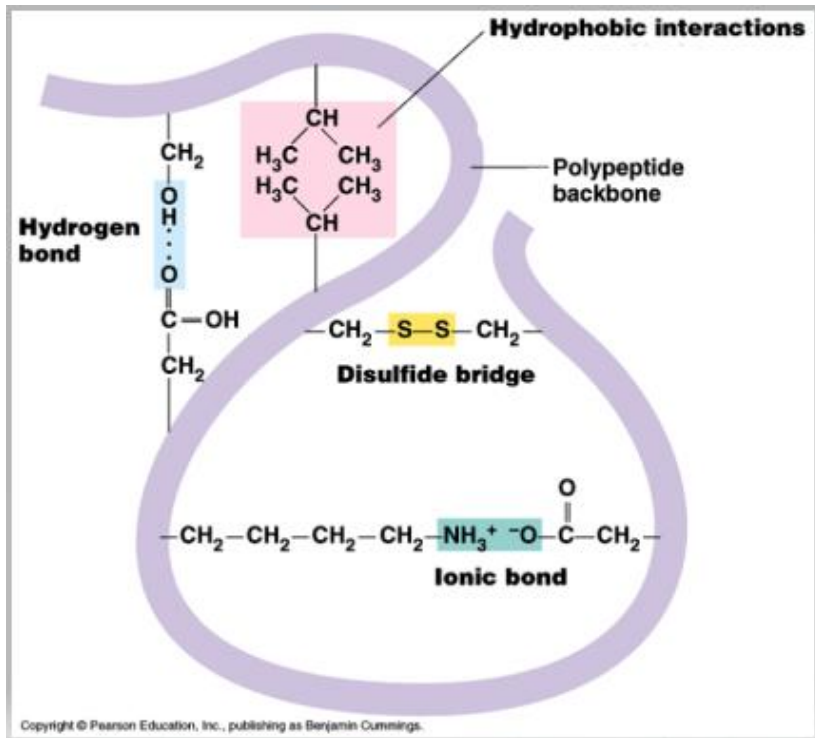
Myoglobin

80% α -helices and no β -sheets

<https://www.rcsb.org/structure/1MBN>

Tertiary Structure of Proteins

- Bending and twisting of the polypeptide chain into compact 3D shape.
- Folding is linked to function; and has to ensure compatibility with environment.

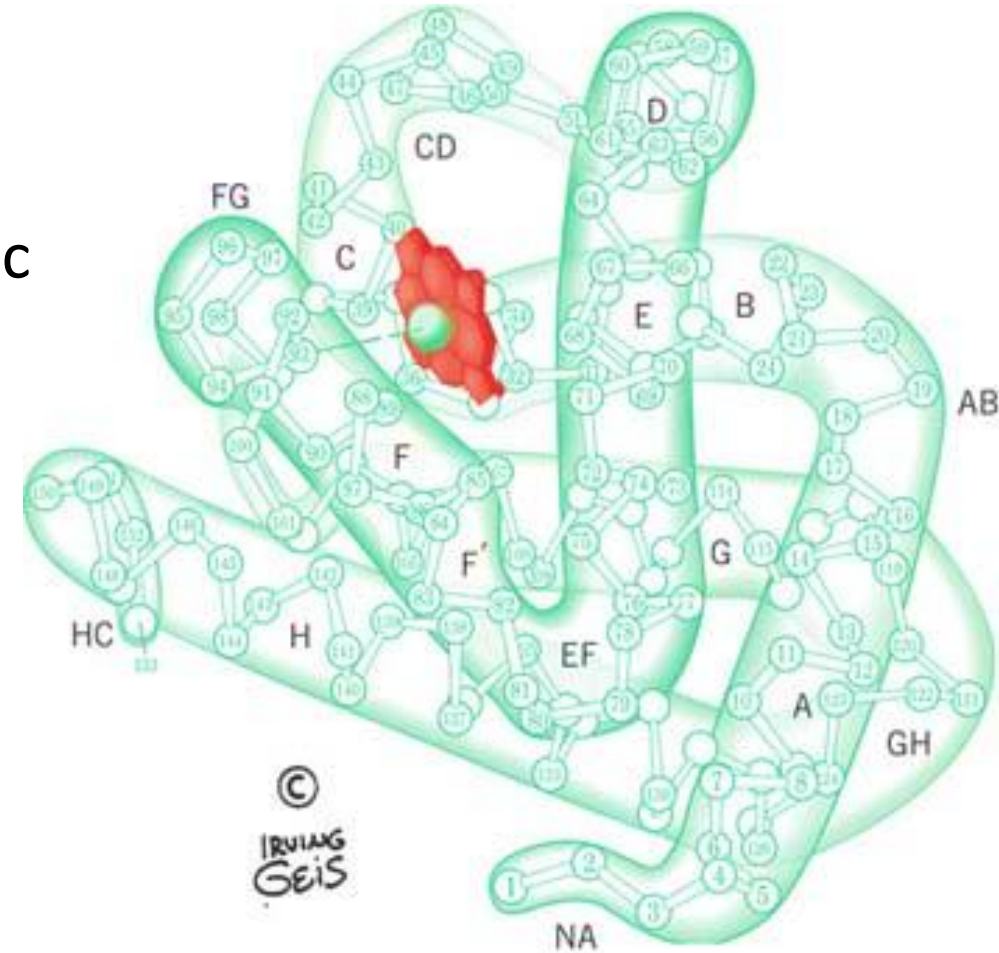


Glyceraldehyde-3-P dehydrogenase

- 2 domains of Glyceraldehyde-3-P dehydrogenase
 - Binds G3P (**orange**)
 - Binds NAD⁺ (**blue**)
- **Domains** - physically independent regions with specific functions.

3° Structure of Myoglobin

- Globular protein
- Located in muscle for storage of O₂
- Single peptide chain (153 amino acids) + prosthetic heme group.
- Structure consists of **8 α-helical regions** (no β-pleated sheets).
- Exterior: Polar amino acid residues (hydrophilic)
- Interior: Non-polar amino acid residues (hydrophobic): except 2 histidines involved in Iron and O₂ binding



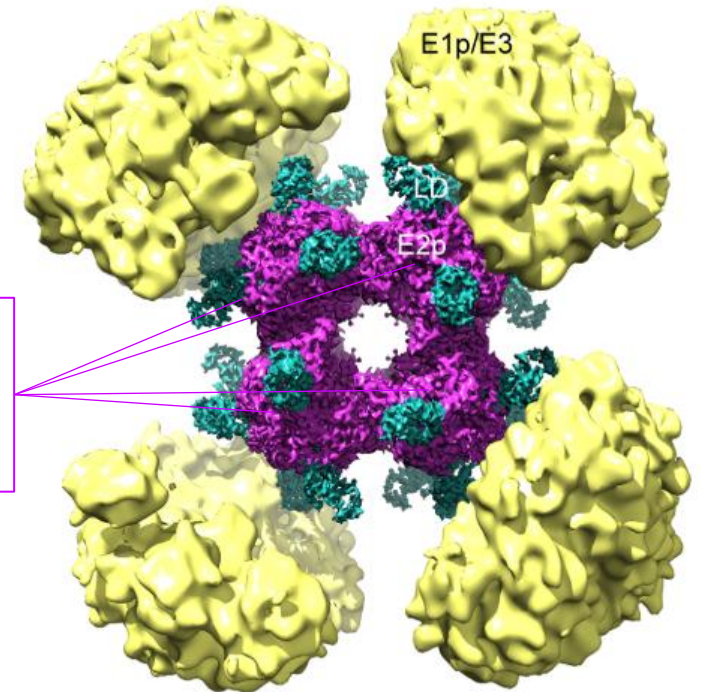
Quaternary Structure (4°) of Proteins

- Association of more than one polypeptide chain.
- Number of chains may vary from 2 (**dimer**), 3 (**trimer**) or 4 (**tetramer**).
- Interactions are primarily **non-covalent** i.e. same as 2° and 3° structure.
- **Homomeric** - polypeptides are homogenous.
- **Heteromeric** - polypeptides are heterogenous.

Purpose:

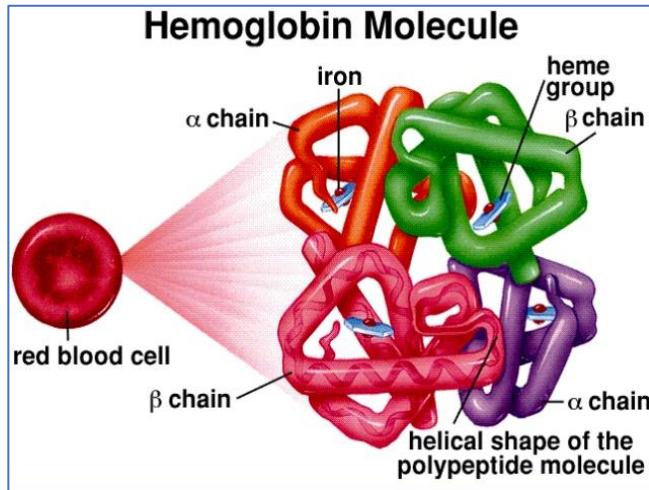
- Increases protein stability and opportunities for cooperativity.
- Brings catalytic sites in close proximity e.g., pyruvate dehydrogenase complex (PDH).

Catalytic cores
of PDH
(magenta)

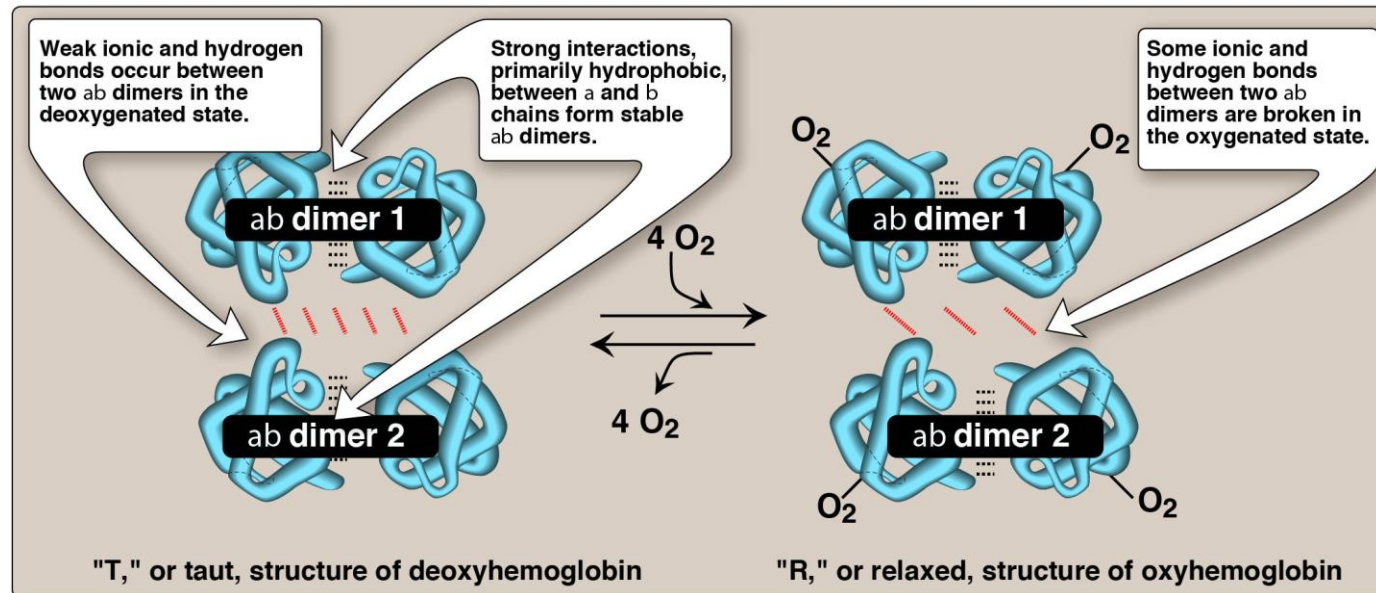


Škerlová, J., et al. Nat Commun 12, 5277 (2021).
<https://doi.org/10.1038/s41467-021-25570-y>

4^o Structure of Hemoglobin (Hb)



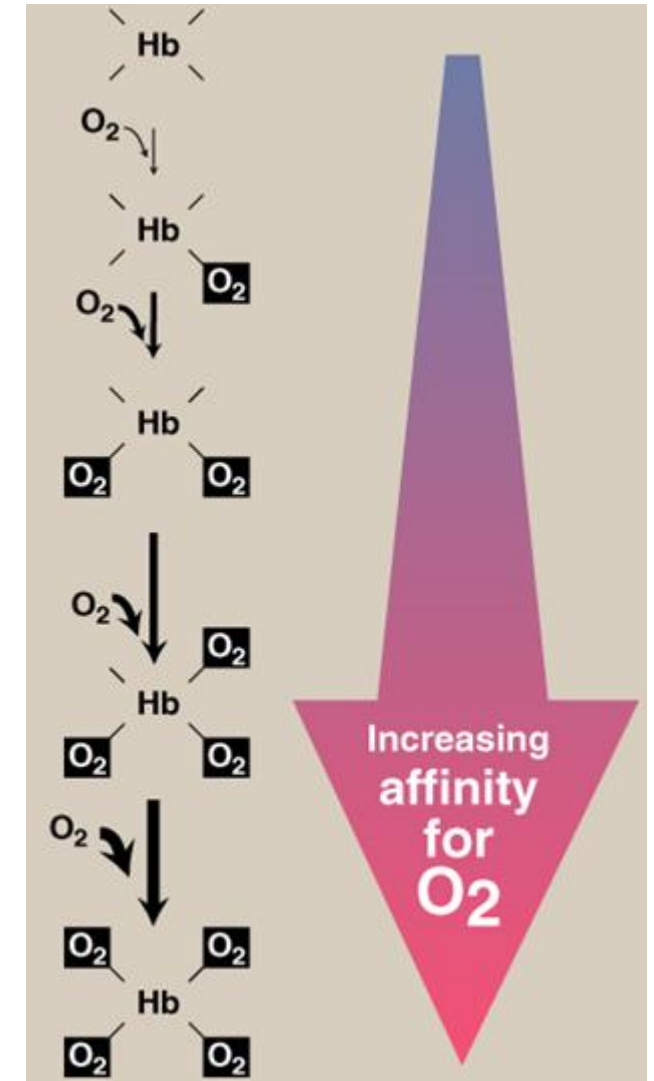
- Contains 2 α and 2 β subunits.
- The 2 α and 2 β subunits form **2 $\alpha\beta$ dimers**. Dimers are held together primarily by hydrophobic interactions and are stable.
- Hb is composed of four polypeptide chains held together by **weak ionic and hydrogen bonds**



Note: These bonds are important for the function of Hb

4^o Structure of Hb Increases Hb Affinity for O₂

- Each Hb subunit binds O₂
- Hb subunits interact **allosterically**
- O₂ binding is **cooperative**
 - Binding of 1st O₂ to one heme group ↑ the O₂ affinity of the remaining 3 in the tetramer.



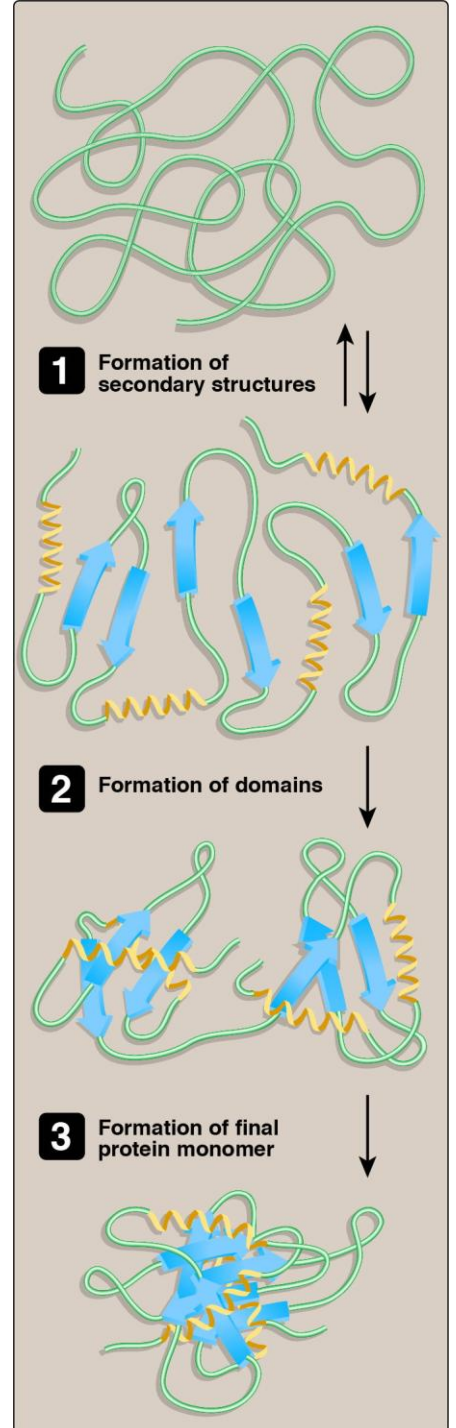
Protein Folding

- Protein synthesized in an unfolded state
- **Must be properly folded to be functionally active.**
- **Folding:** Orderly sequential process (non-random) by which proteins attain lowest possible state of energy.

2° structure is organized. Polarity of the side chains facilitate formation of 2° structures (hydrophobic effect) (*hydrophobic groups come together as water is released*).



3° structure forms & stabilized by non-covalent interactions.

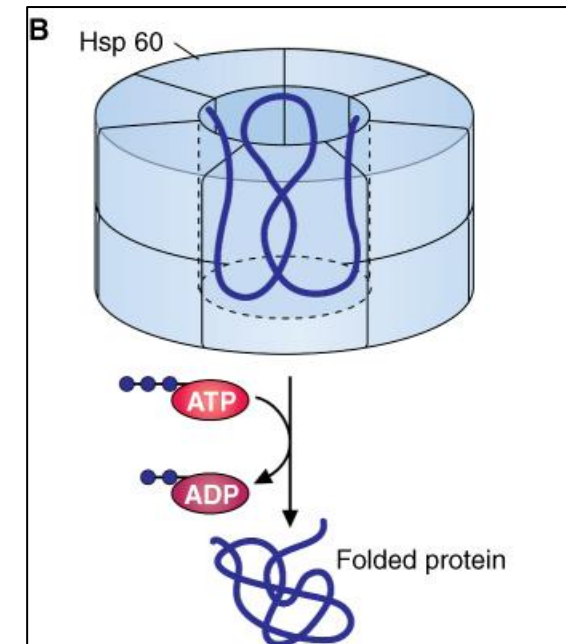
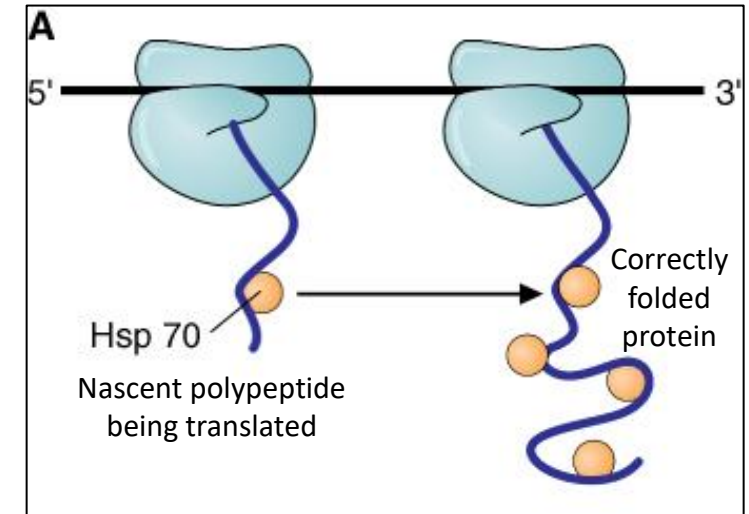


Role of Molecular Chaperones in Protein Folding

- **Molecular chaperones** e.g., Hsp, GroEL/GroES complex facilitate folding
 - Assist/ensure correct folding, accelerate process, direct proteins for degradation...
 - Require ATP hydrolysis and are often associated with ribosomes

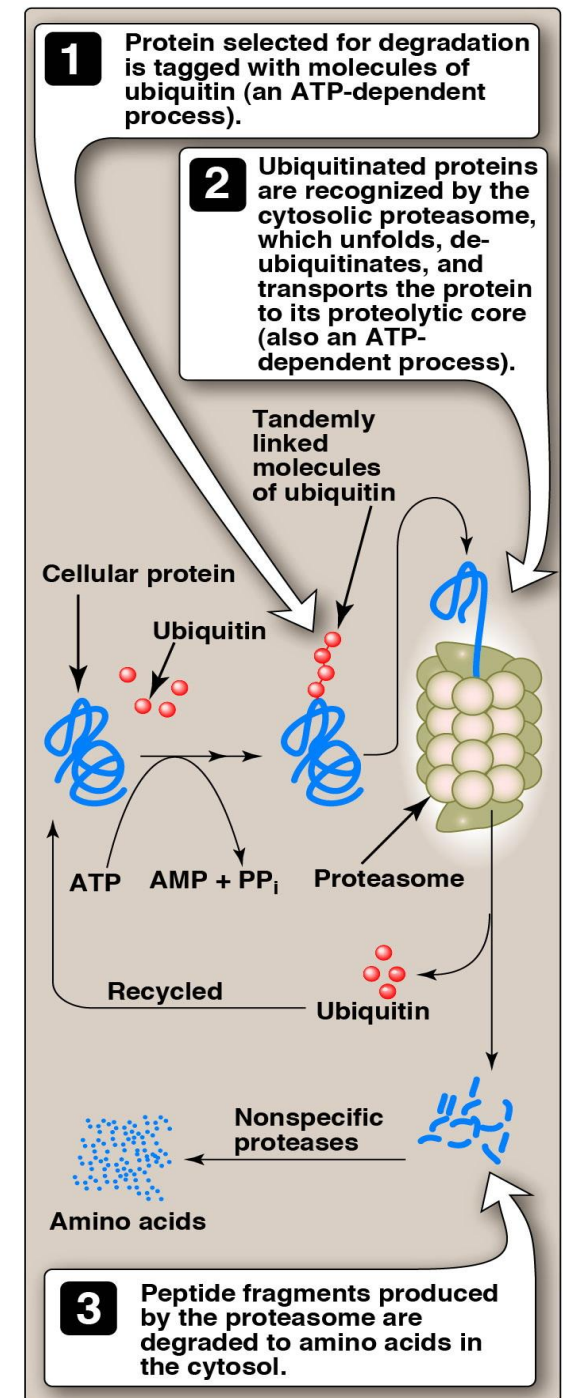
Functions of Chaperones: Hsp 70 and Hsp 60 Family Members

- **Hsp 70** - prevents aggregation of unfolded protein.
 - Binds to hydrophobic regions of an extended polypeptide and prevents premature protein folding
- **Hsp 60** - required for correct folding of cellular proteins that do not fold spontaneously.



Protein Turnover: Ubiquitin-Proteasome Complex

- Misfolded or defective proteins are normally tagged by the small protein **ubiquitin** and degraded in the **ubiquitin-proteasome system**.
- The linkage of ubiquitin is an ATP-dependent process.
- The ubiquitinated proteins are degraded in the cytosolic proteasome and ubiquitin is recycled.
- The proteasome unfolds and cuts the target protein into fragments that are further degraded by cytosolic proteases.



Protein Structure Abnormalities and Disease

Some examples that we will consider:

- Hemoglobinopathies (HbS)
- Glycated Hemoglobin
- Glycation & oxidation of Collagen
- Alzheimer disease
- Prion disease
- Protein denaturation

Remember Serum Proteins are also used as diagnostic tools (see CPR module 😊)

Hemoglobinopathies

- Definition: a group of genetic diseases whereby synthesis of **structurally abnormal Hb** or insufficient amounts of normal Hb (or rarely both) occur.
- Examples: Sickle cell anemia (HbSS), hemoglobin SC disease (HbSC), the Thalassemias
- May have serious clinical implications

Autosomal Recessive Sickle Cell Anemia (HbSS)

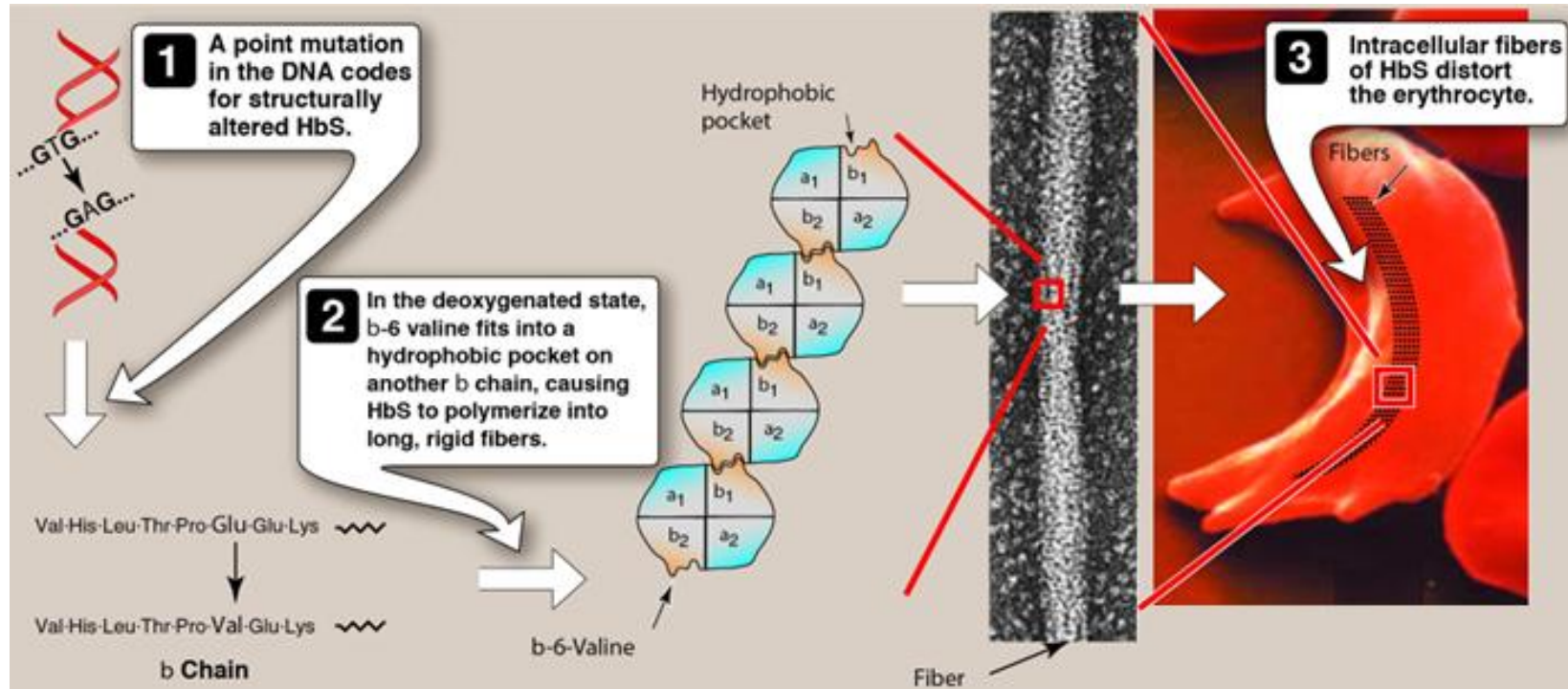
T→A Point mutation

HbS: Glu6Val

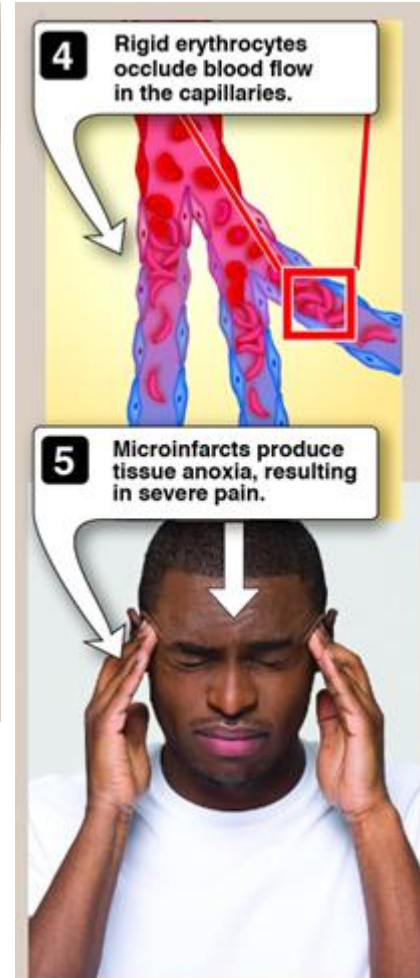
Side chains:

Glu = polar

Val = nonpolar hydrophobic



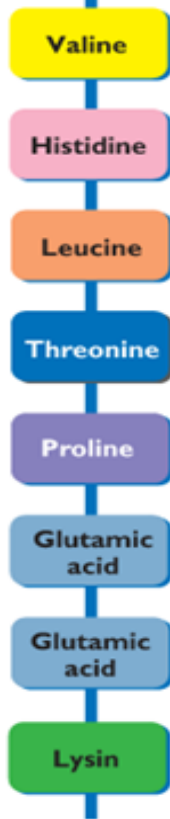
- Glu6Val substitution → **polymerization of deoxygenated HbS in RBCs which stiffens and distort RBC into a sickle shape.**
- Sickled RBCs die prematurely, leading to shortage of RBCs (**anemia**).
- Sickled RBCs can also block small blood vessels (infarcts), causing pain and ischemic death of nearby cells or organ damage.



Hemoglobin Electrophoresis from Lysed RBC

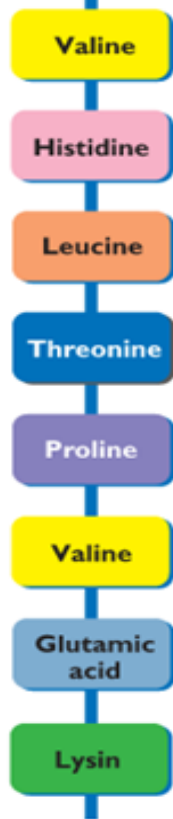
HbA

Normal Hemoglobin



HbS

Sickle Cell Hemoglobin

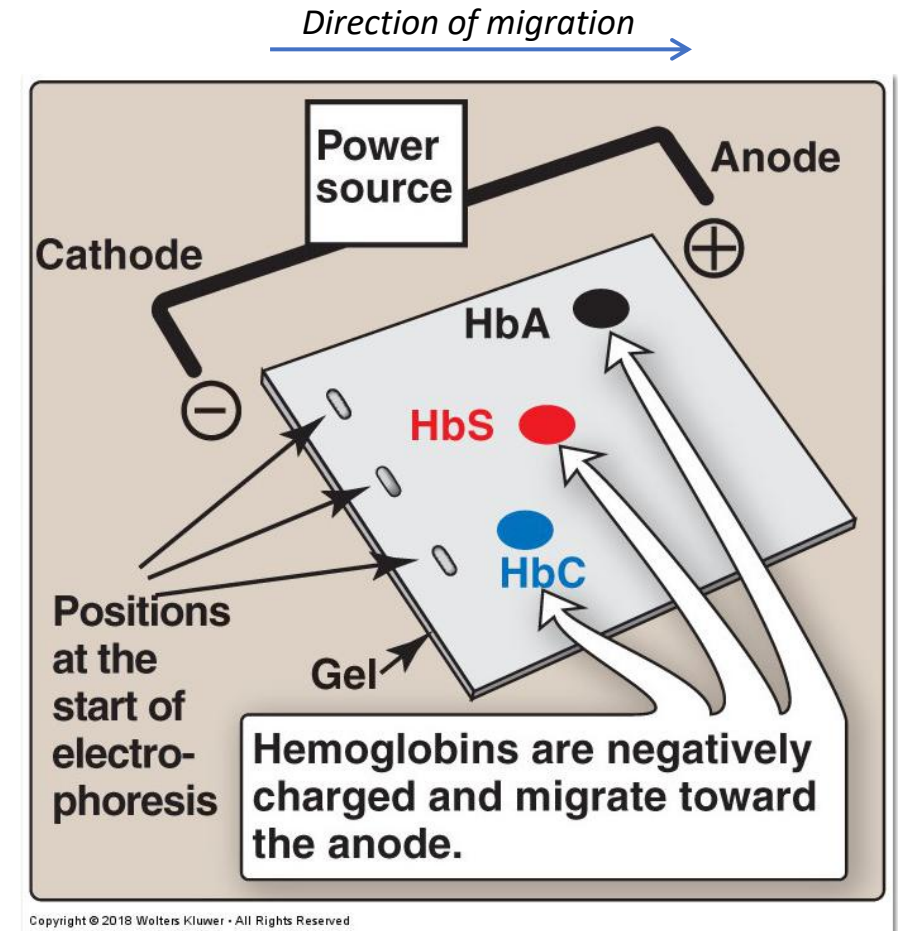


The only structural difference is one change in the sequence of each beta chain.

- ve

HbS: **Glu6Val**

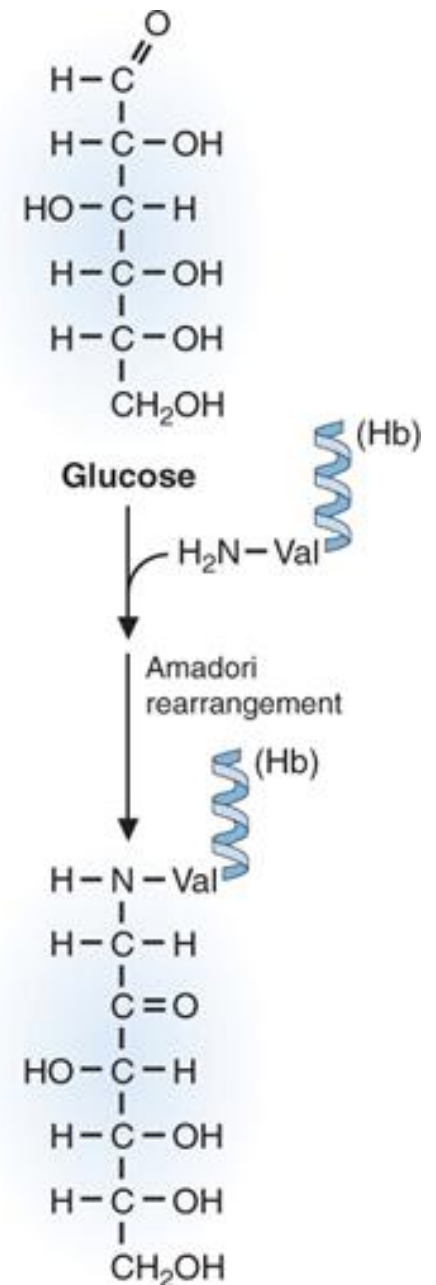
➤ **Mobility of HbS is altered during agarose gel electrophoresis at alkaline pH (*Hb assumes negative charge at this pH*).**



HbS is less negative than HbA due to loss of Glu → does not migrate as far as normal HbA

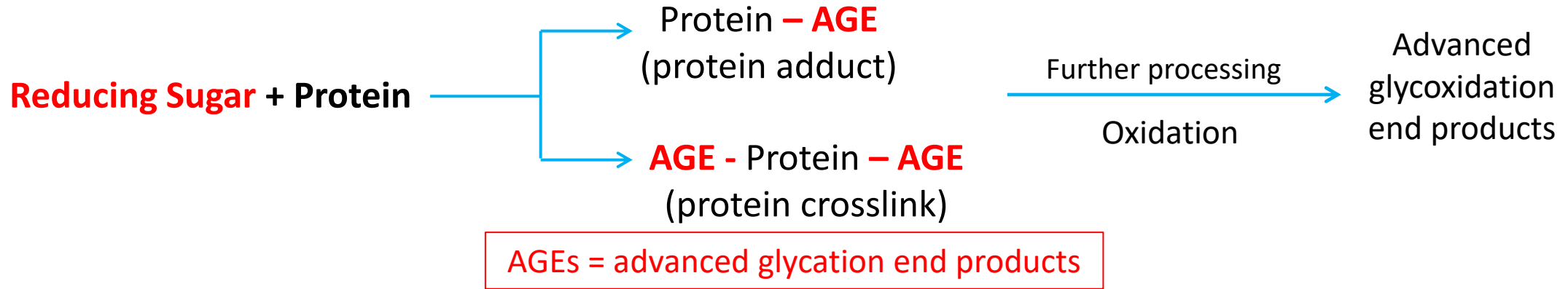
Glycated Hb

- Glycation of Hb (nonenzymatic addition of glucose) can occur as a result of hyperglycemia.
- Glycated hemoglobin (HbA1c) level is used as a biomarker in the diagnosis and treatment of diabetes and as an **indicator for long-term control of glycemic state**.
- Estimate HbA1c levels by measuring degree of modification of hemoglobin.



Glycation and Oxidation of Collagen

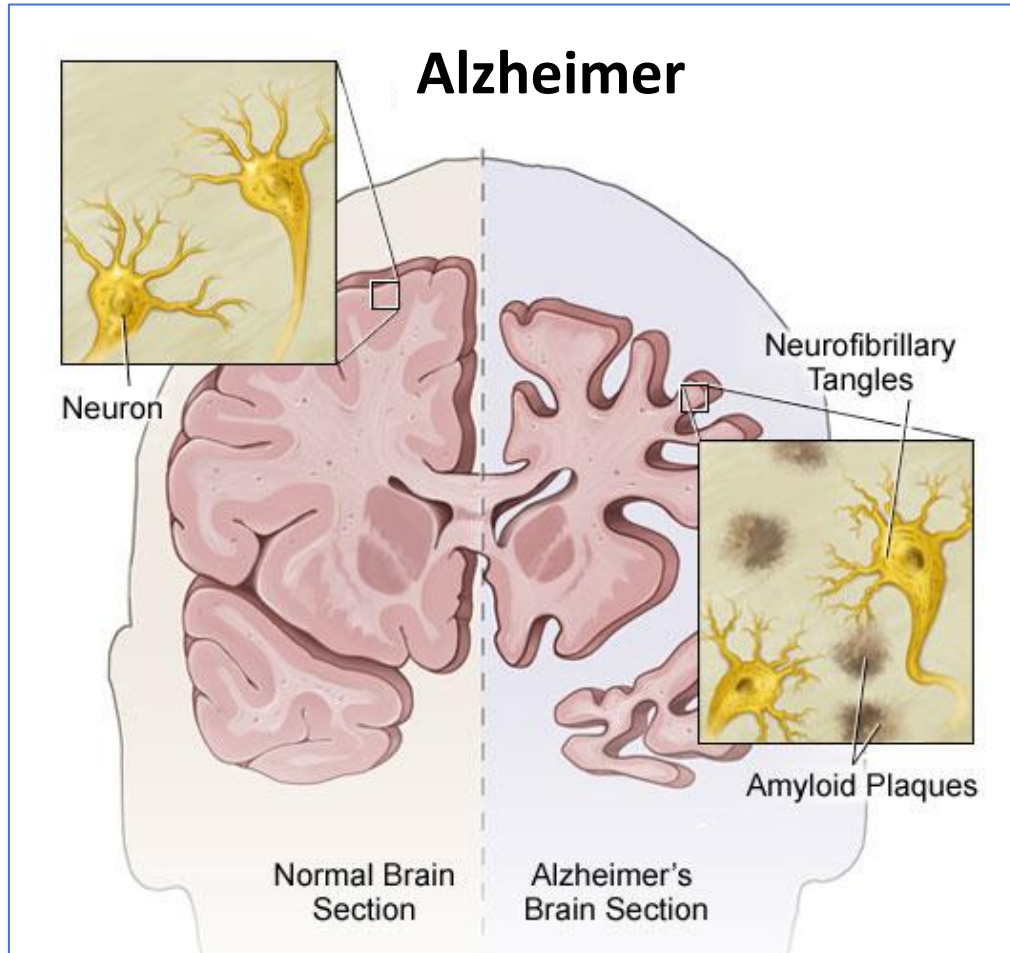
- **Non-enzymatic** modification of proteins e.g. Glycation or oxidation of collagen.



- Glycation initially affects the interactions of collagen with cells and other matrix components.
- The most damaging effects are caused by the formation of glucose-mediated intermolecular cross-links.
- These cross-links decrease the critical flexibility and permeability of the tissues and reduce turnover.

Pathologic Aggregation of Proteins (Deposits of Misfolded Proteins)

Disease can occur when an apparently normal protein assumes a conformation that is cytotoxic: **Alzheimer and Prion disease**



How Creutzfeldt-Jakob disease works

CAUSE

Creutzfeldt-Jakob disease is caused by abnormal proteins called prions that are not killed by standard methods for sterilizing surgical equipment.

CONSEQUENCES

Those affected lose the ability to think and to move properly and suffer from memory loss. It is always fatal, usually within one year of onset of illness.

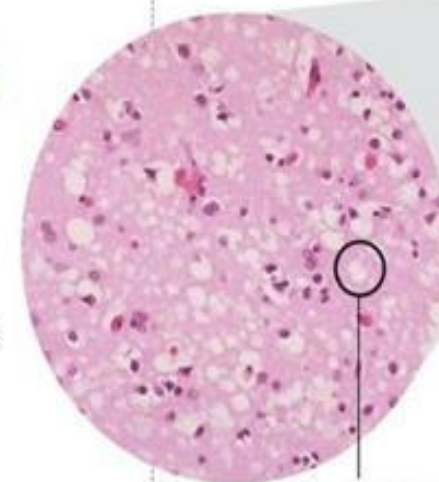


**NORMAL
HUMAN
PROTEIN**

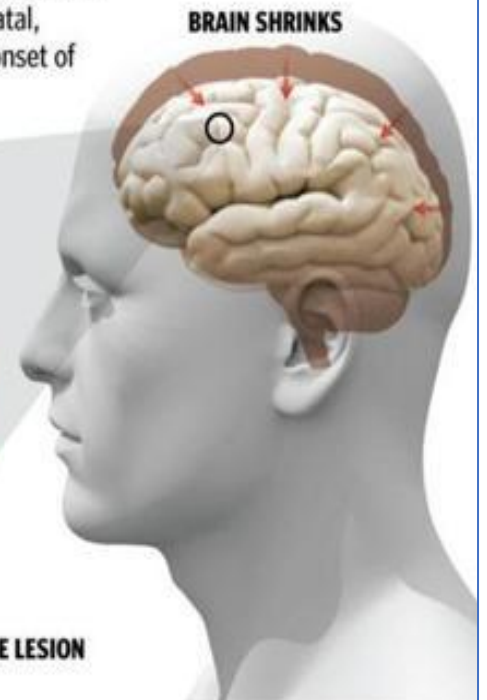


**DISEASE-
CAUSING
PRION**

As prions build up in cells, the brain slowly shrinks and the tissue fills with holes until it resembles a sponge.



SPONGE-LIKE LESION



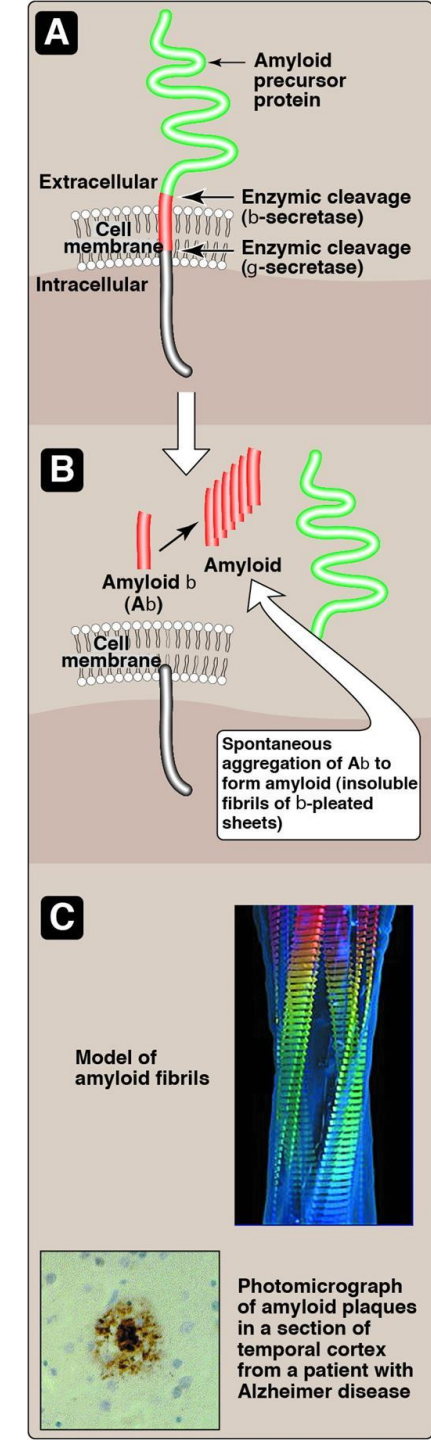
BRAIN SHRINKS

SOURCES: World Health Organization, Centers for Disease Control and Prevention, National Institute of Neurological Disorders and Stroke, AP

DAVID BUTLER, CHIQUI ESTEBAN, JAVIER ZARRACINA/GLOBE STAFF

Alzheimer Disease

- Neurodegenerative disorder; Affects older adults
- Impairment in memory and cognitive decline
- Abnormal proteolytic cleavage of **amyloid precursor protein** → spontaneous formation of fibrillar protein aggregates known as **beta-amyloid plaques**
- **Amyloid fibrils** are neurotoxic → Cognitive impairment
- Also, accumulation of neurofibrillary tangles in neurons
 - Contain **hyperphosphorylated tau protein**
 - Normal function of tau: Assist microtubule assembly



Prion diseases: Transmissible Spongiform Encephalopathy (TSE)

Group neurodegenerative disorders characterised by rapid progressive dementia & ataxia.

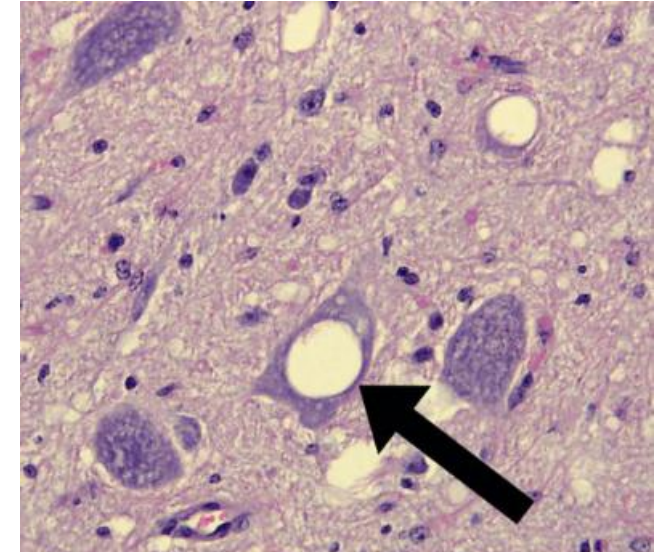
Pathophysiology - accumulation of fibrillary or insoluble prion proteins



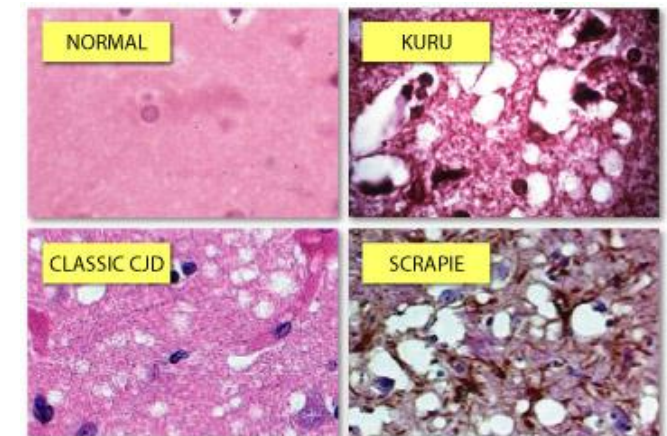
Degeneration of neurons & vacuolisation termed SPONGIFORM DEGENERATION

Includes:

- Classic Creutzfeldt Jakob's Disease (CJD) – most cases sporadic (unknown etiology)
 - Variant CJD – linked to consuming BSE infected beef
 - Kuru – linked to cannibalism - New Guinea 1950s (now rare)
-
- Bovine Spongiform Encephalopathy (BSE) a.k.a. “Mad-cow” disease (1996)
 - Scrapie in sheep

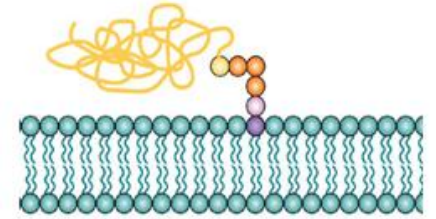


Spongiform change in the cerebral cortex - note the optically empty vacuoles on the neuronal cell body (black arrow)



Abnormal Prions: a Proteinaceous Infectious Particle

- *PRNP* gene encodes the **NORMAL** prion protein (PrP or PrP^c).
- Highest level of expression in neurons
- Encodes a cell-surface glycoprotein linked to the cell membrane

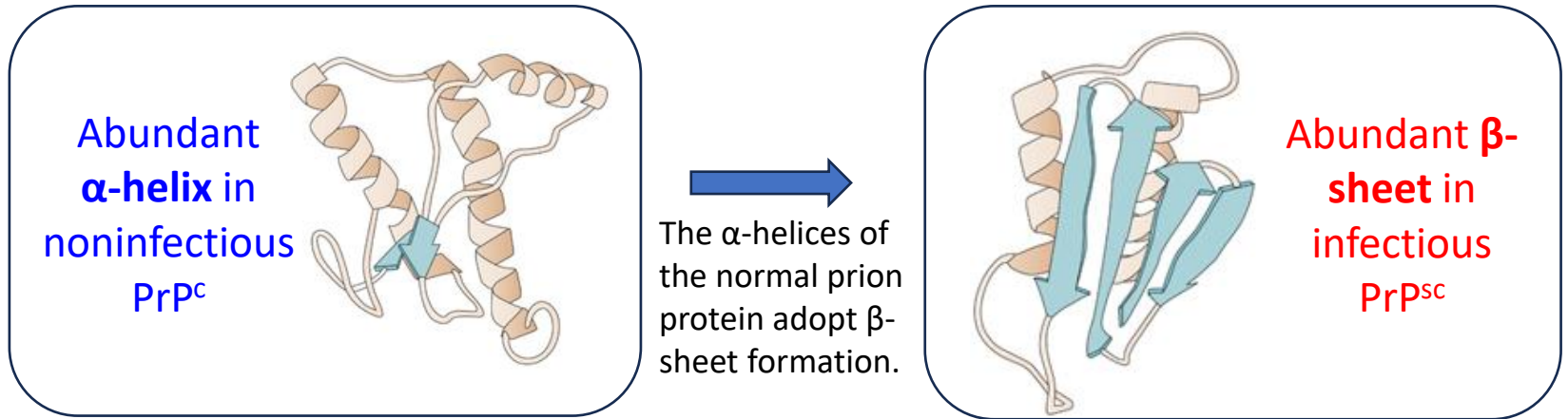


Pathogenic PrP called 'scrapie PrP' or PrP^{sc}



- **No 1^o structure differences** or alternate translational modifications of the disease-causing form of the prion protein has been discovered.
- The key of becoming infectious results from **changes in the 3D conformation** due to **changes in 2^o structures**.

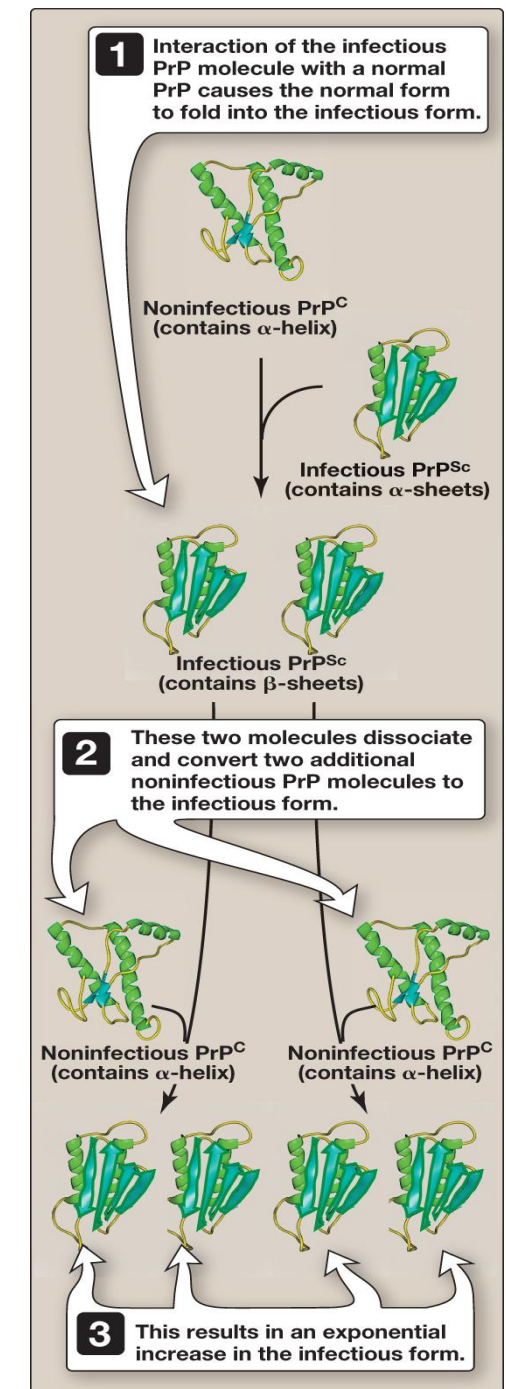
Changes in 2^o Structure of the Infectious Prion



✓ The infectious prion has **more β -sheets**.

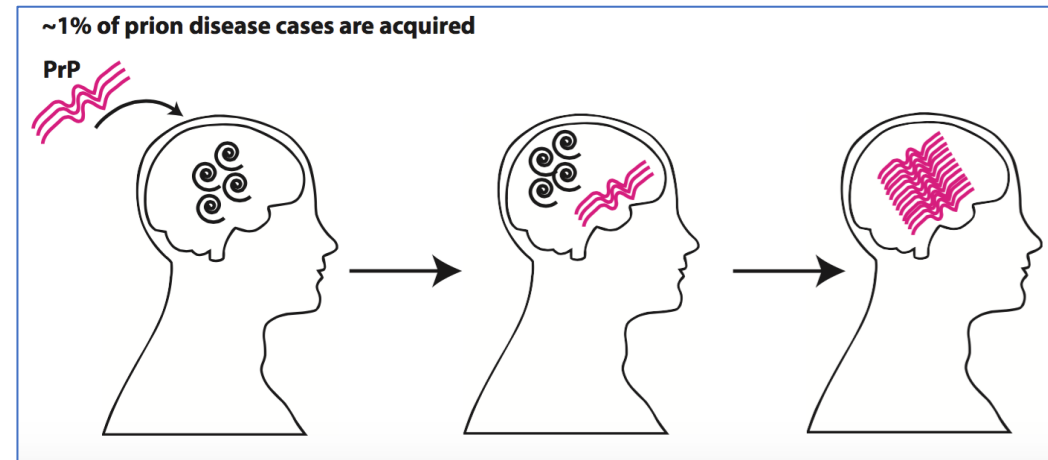
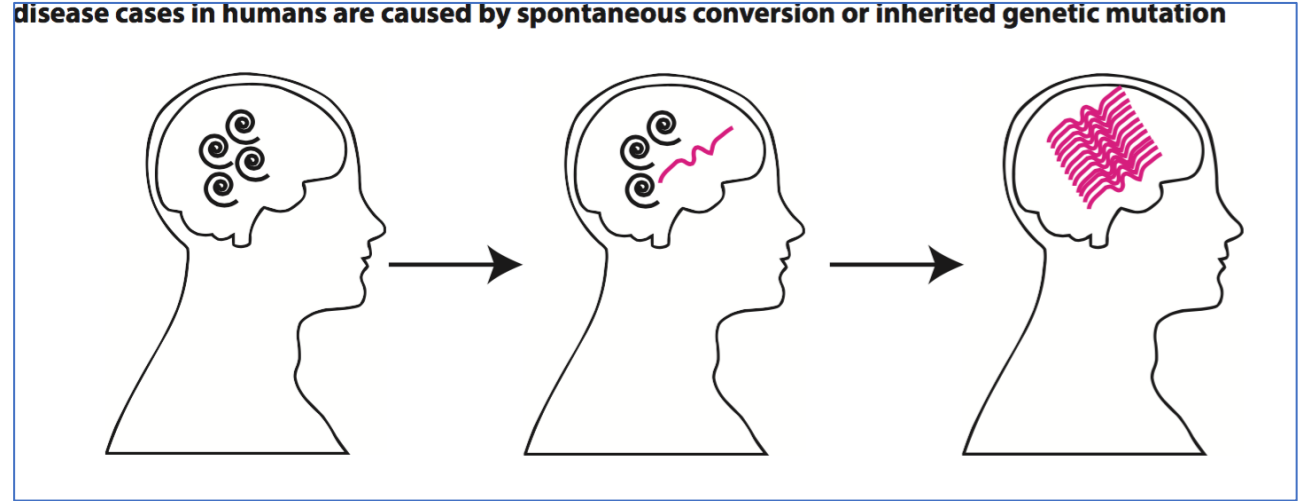
PrP^{Sc}: Scrapie PrP

- Acts like a template and leads to conversion of PrP to large aggregates of **PrP^{Sc}** (*autocatalytic and exponentially expands in presence of abnormal PrP^{Sc}*) → **Prion disease**
- Not destroyed by ubiquitin-proteasome complex or by sterilization of surgical instruments; Highly infective
 - Require protein denaturing agents (high heat/ alkali) for decontamination



Routes of Transmission of Prion Disease

- Sporadic (most cases)
- Genetic predisposition (mutations)
- Acquired:
 - Ingestion of contaminated meat (brain)
 - Blood transfusion
 - Invasive brain surgeries and procedures
 - Injection of contaminated human growth hormone.



Protein Denaturation

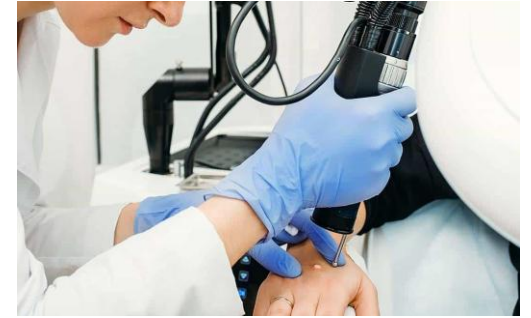
- **Non-covalent interactions in proteins are numerous but weak.**
- Easily disrupted → unfolding of the protein (**known as denaturation**) → **Generally results in loss of function of protein**
- Denaturation breaks down 2° , 3° and 4° structure of proteins.
- **Peptide bonds only break under extreme conditions:** e.g., prolonged exposure to strong acid and/or extremes of temperature .
- Denaturation is generally **irreversible**

Denaturation of proteins is often used as a means of **disinfection**.



Protein Denaturation *In Vivo*

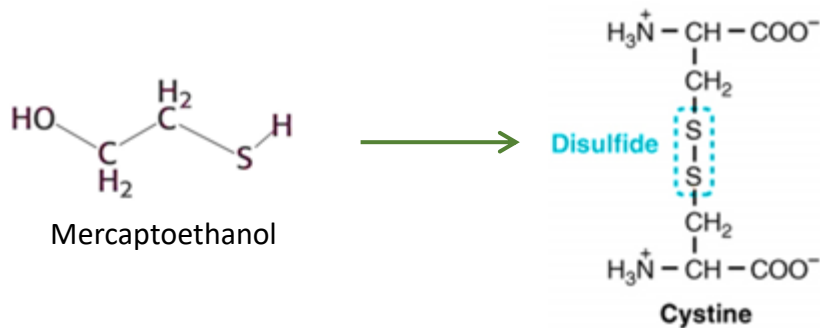
- **Cryosurgery** involves the removal of tissue using **very low temperature (freezing)** which results in protein denaturation and cellular death.
- **Electrosurgery** involves use of **electric current** (generates **heat**) to result in denaturation and cell death
 - Used to reduce bleeding during surgery
- Dietary proteins are denatured by the gastric juice at low pH → **Facilitates opening of the native structure**
→ **Facilitates action of digestive enzymes.**



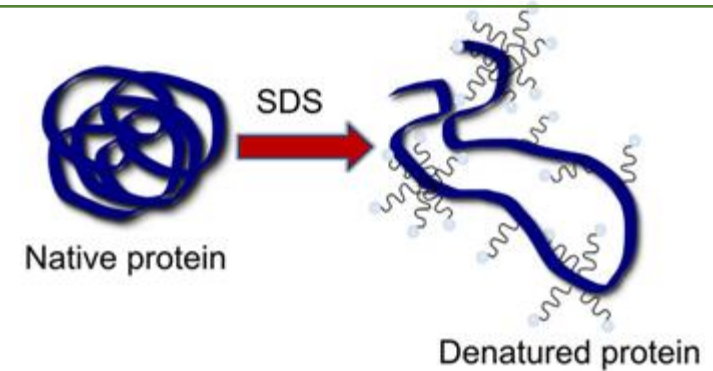
Protein Denaturation in the Laboratory

Proteins can be denatured by:

- **Heat** (usually $>60 - 70^{\circ}\text{C}$) – increases the vibrational & rotational energies of bonds.
- **Extremes in pH:** $\downarrow$ pH disrupts bonds involving COO^- groups; $\uparrow$ pH disrupts bonds involving NH_3^+ groups



- **Reduction of disulfide bonds**
e.g., mercaptoethanol & other reducing agents

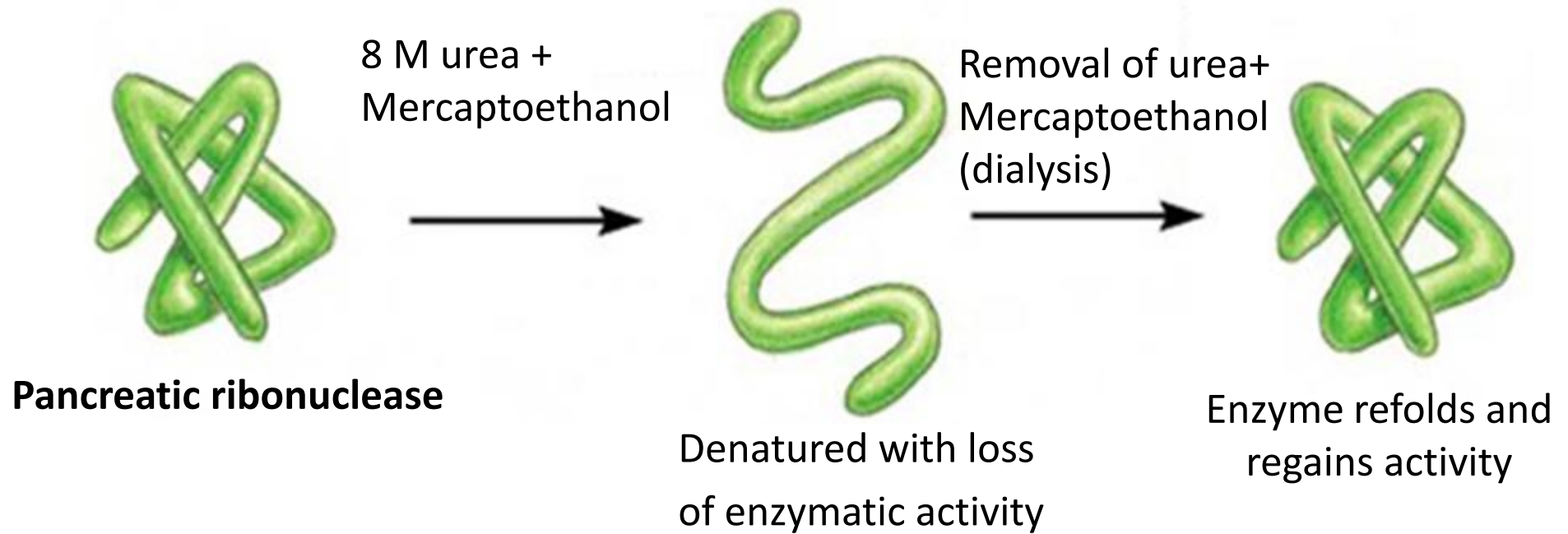


- **Detergents** disrupt hydrophobic interactions
e.g., sodium dodecyl sulphate (SDS).

- **Organic solvents:** Disruption of aqueous environment and thus hydrophobic interactions.
- **Chaotropic:** Urea/guanidinium have a strong capacity to form H bonds (Disrupt secondary structure).

Protein denaturation generally irreversible

Exceptions to the rule:



- Thermophilic bacteria contain very heat stable proteins → Used for Molecular biology techniques (DNA polymerase used for PCR)

Stabilizing Bonds (Forces) Found in Proteins

Self-study

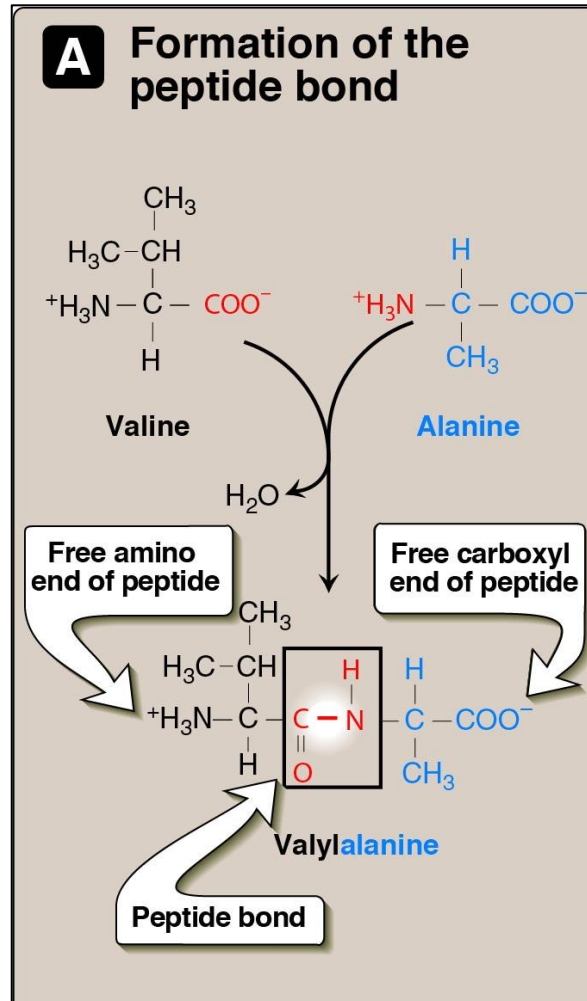
(Students are expected to review the information for better understanding)

Covalent Bonds

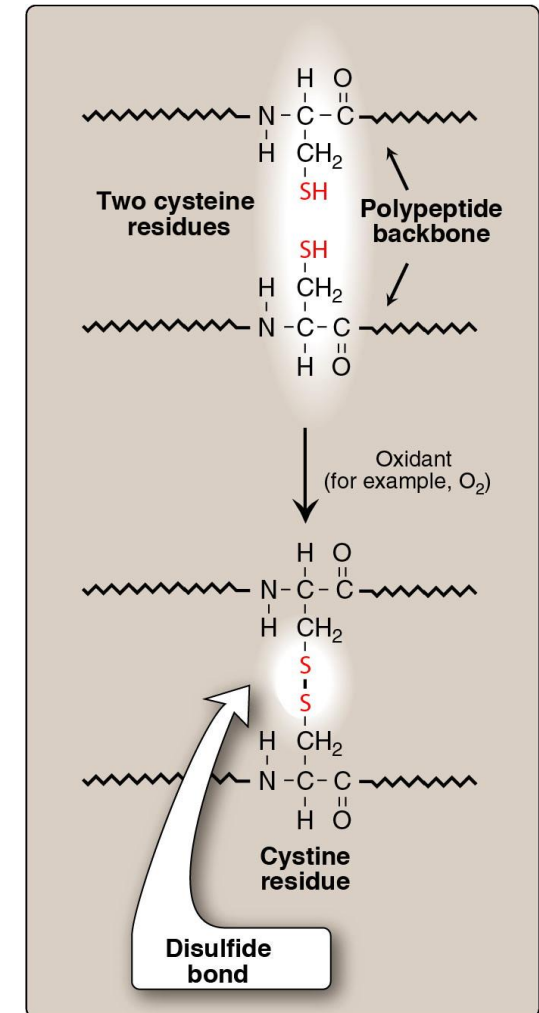
Self-study

Covalent bonds are strong

- Examples: **peptide bonds** and **disulfide bonds** are enzymatically formed.
- Generally, not broken other than during protein degradation



Peptide bond formed by two amino acids

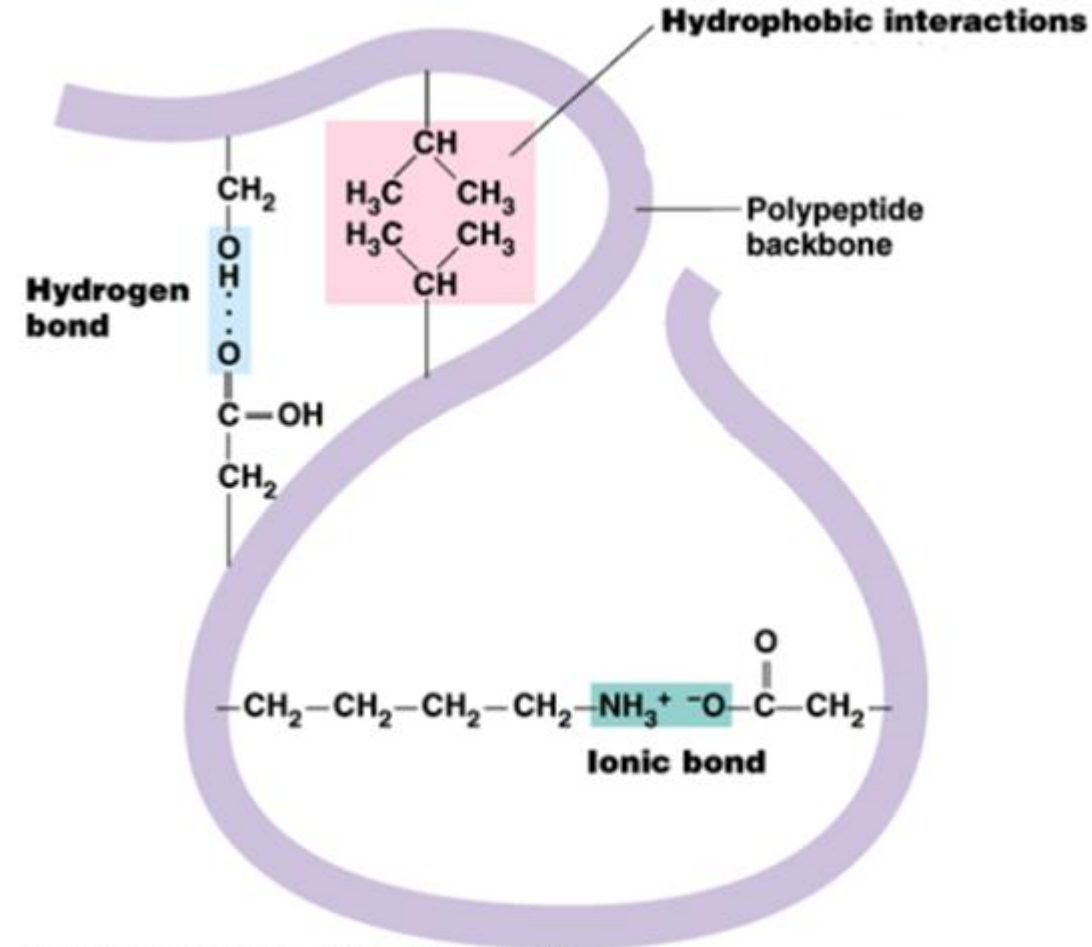


Disulfide bond formed by two **cysteine** residues

3^o Structure includes Covalent and Noncovalent Bonds

Self-study

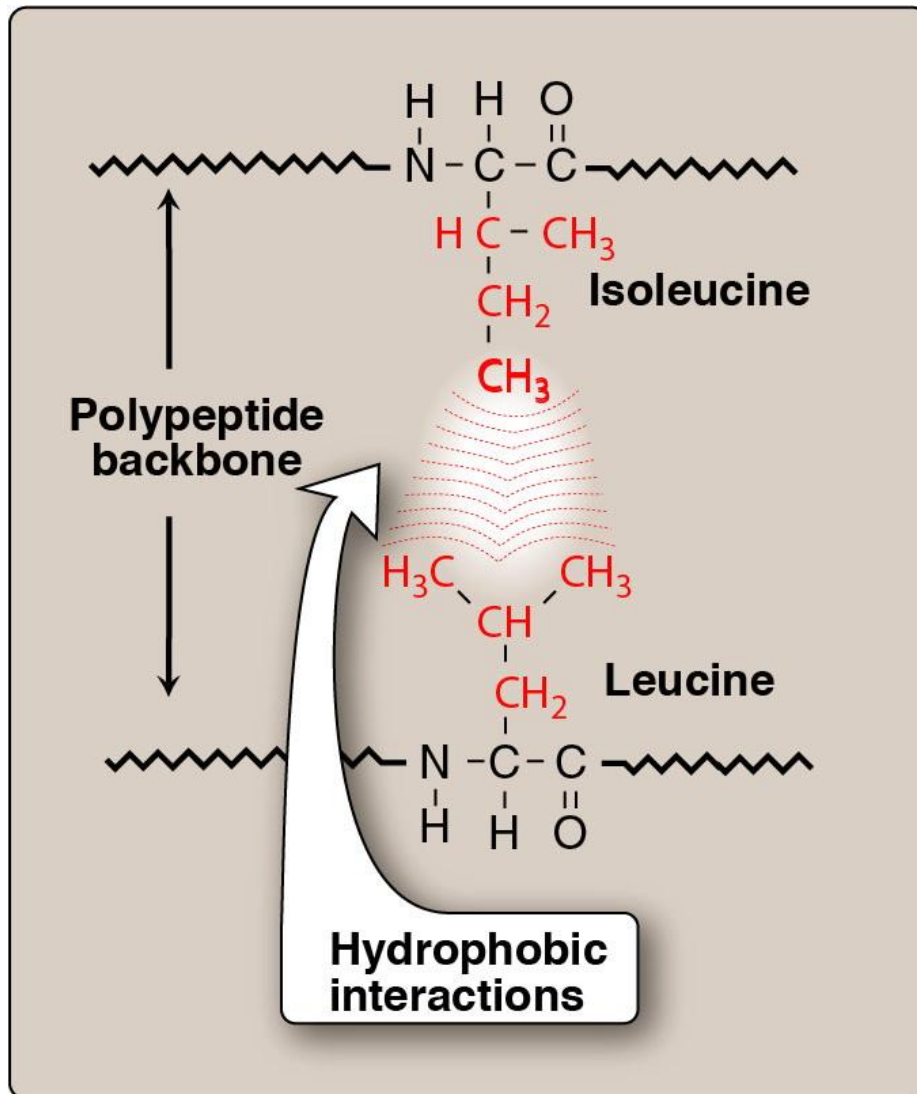
- **Noncovalent bonds** are weaker than covalent bonds.
- These “weak attractive forces” specify protein folding and conformational changes & include:
 - Hydrophobic interactions
 - Hydrogen bonds
 - Ionic bonds
- These forces may be additive, so in aggregates they may provide a strong stabilizing force.



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Hydrophobic Interactions (Water Exclusion)

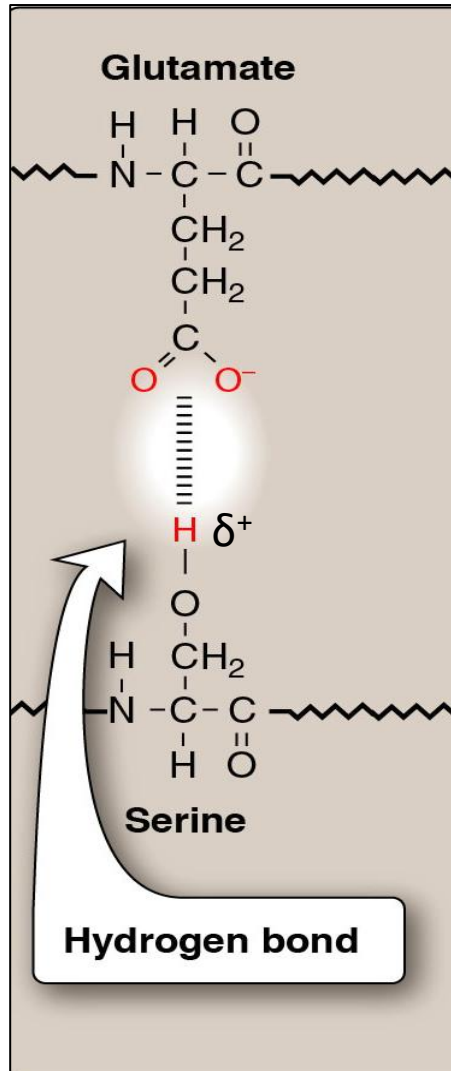
Self-study



- Amino acids with nonpolar side chains can associate with other hydrophobic amino acids.
- Nonpolar amino acids tend to be located in the interior of the folded protein.
- Water is excluded.
- The **branched-chain amino** acids (valine, leucine, isoleucine), **alanine** and **phenylalanine** are commonly found to participate in **hydrophobic** interactions.

Hydrogen Bonds (A Hydrogen Atom is Shared)

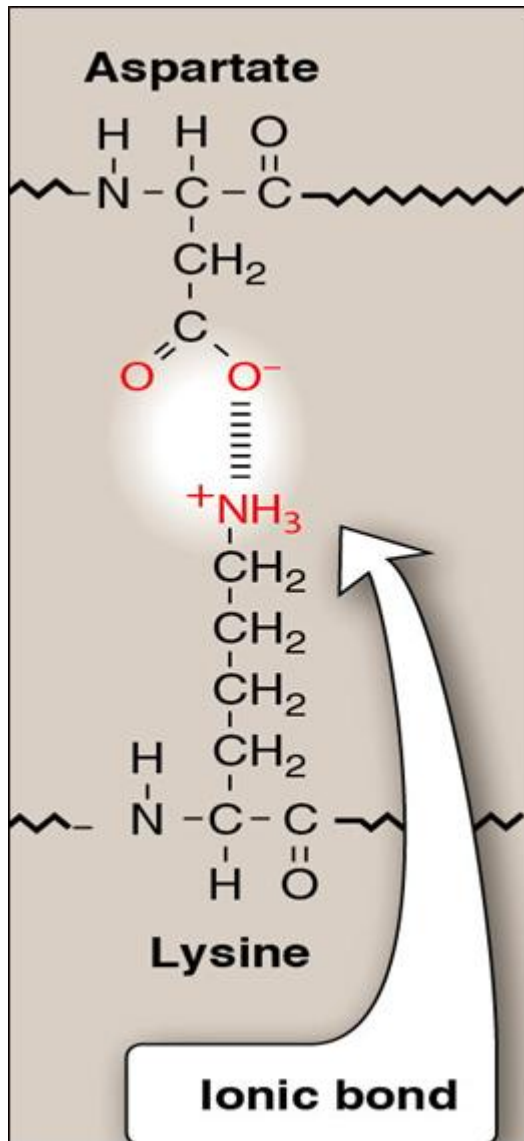
Self-study



- A hydrogen donor ($-\text{OH}$ or $-\text{NH}$) must interact with a hydrogen acceptor ($-\text{O}$ or $-\text{N}$).
- Hydrogen bonds **between peptide bonds** stabilize the secondary structures of α -helix and β -sheet.
- Hydrogen bonds **between amino acid side chains** stabilize the tertiary structure of proteins.
- Example: the **serine** side chain can form a hydrogen bond with
 - the charged side chain of **glutamate or aspartate**
 - the uncharged polar side chain of an **asparagine or glutamine**.

Ionic Bonds (Electrostatic Attraction)

Self-study



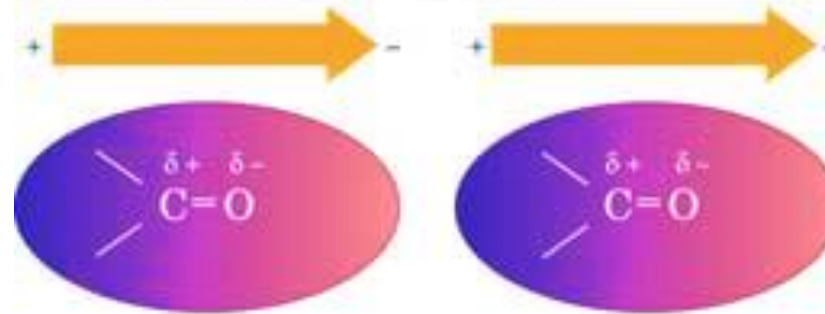
- **Ionic bonds** are formed between negatively and positively charged side chains of amino acid residues
- **Negatively charged side chains** are found in aspartate and glutamate residues
- **Positively charged side chains** are found in lysine, arginine and histidine residues

Van der Waals Forces

Self-study

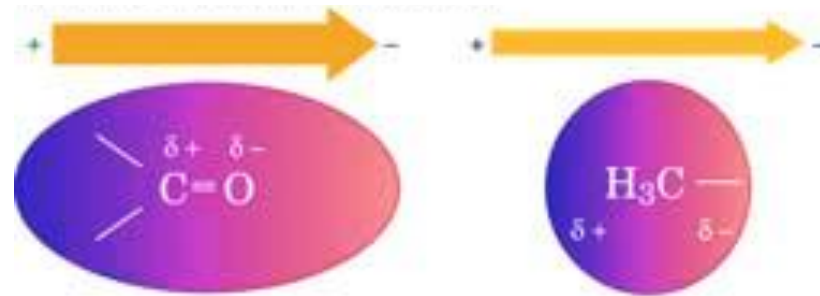
- Weakest bonds

a) Interactions between permanent dipoles

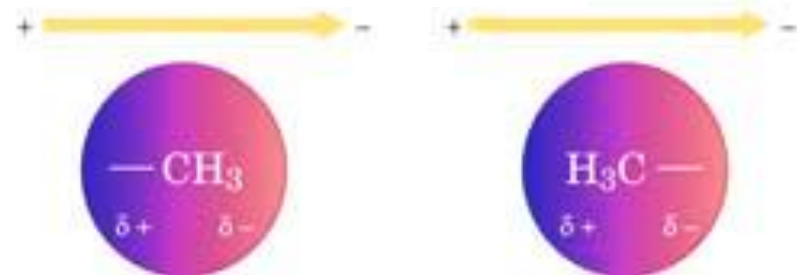


Asparagine, Glutamine

b) Dipole-induced dipole interactions



c) London dispersion forces



Hydrophobic amino acids

Thank you.

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